

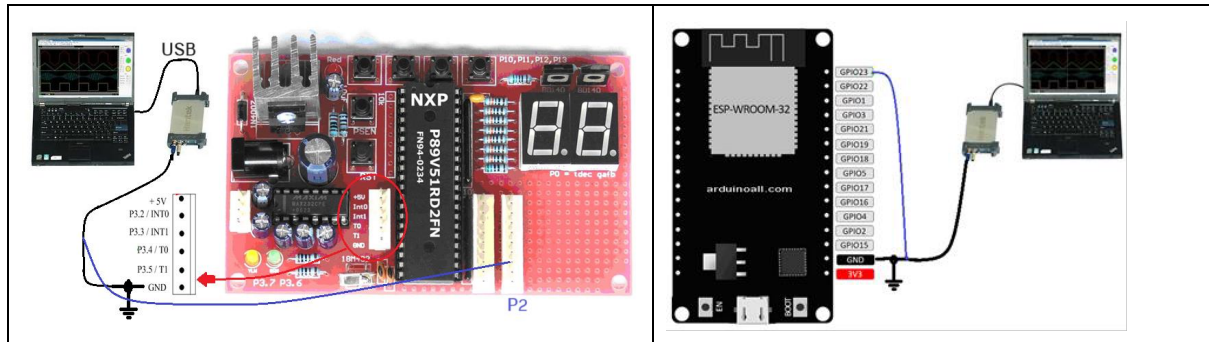
# Week09 – Lab08: Timer, Counter and Timmer Interrupts – ESP32 RTOS

PC \_\_\_\_ รหัส \_\_\_\_ ชื่อ-สกุล \_\_\_\_

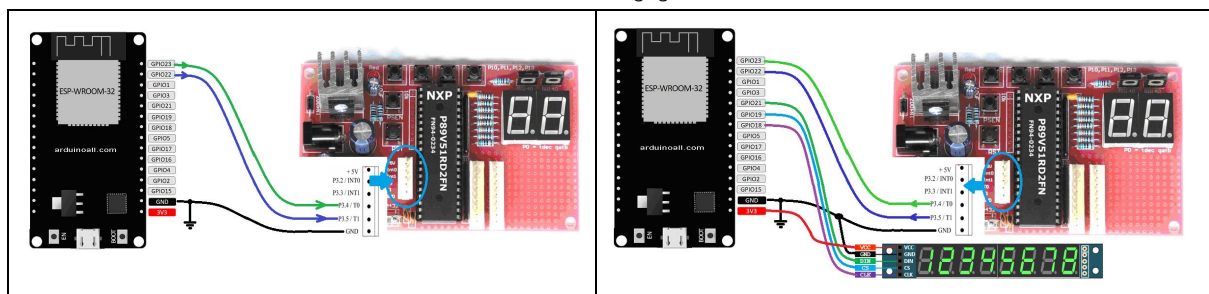
เปลี่ยนชื่อไฟล์เป็น “B6501234-นายไมโคร รายงานแลป-Lab08.pdf”

ส่งงาน Report ที่ >> <https://forms.gle/2jCaarVZxfR2DMbP7>

1. MCS-51 (A) Timer Delay >> สร้างสัญญาณ จำนวน, เวลาOn, เวลาOff
2. ESP-32 (A) Timer Delay >> สร้างสัญญาณ จำนวน, เวลาOn, เวลาOff



3. MCS-51 (B) Counter with Software >> อ่านสัญญาณ จำนวน, เวลาOn, เวลาOff
4. ESP-32 (B) Counter with Software >> อ่านสัญญาณ จำนวน, เวลาOn, เวลาOff

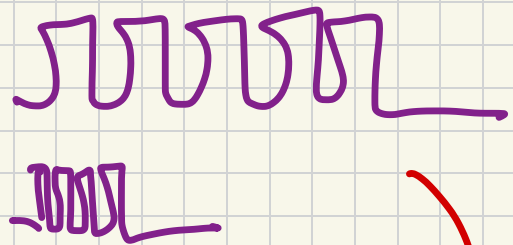
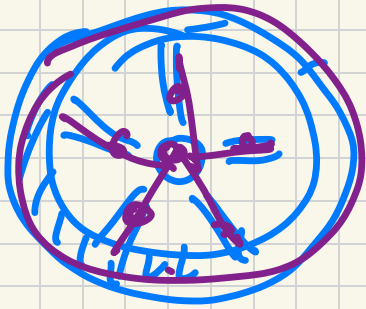


5. Node-RED (A)
  - จาก Mission-5: From ESP-32 to MQTT Broker and If xxx then LINE (or Telegram) Alert
  - หากอุณหภูมิ เกิน xxxx หรือ ความชื้น เกิน yyyy ให้มีการแจ้งเตือนผ่าน LINE (or Telegram)
6. Node-RED (B)
  - จาก Mission-6: From ESP-32 to MQTT Broker and Show on Node-RED Dashboard
  - ให้แสดงอุณหภูมิและความชื้นจาก DHT-22 ที่ Node-RED Dashboard
  - และสามารถควบคุม LED ได้ 2 ตัว

| MCS-51 Timer | ESP-32 Timer | MCS-51 Counter | ESP-32 Counter | Node-RED<br>If xxx then LINE | Node-RED<br>Dashboard |
|--------------|--------------|----------------|----------------|------------------------------|-----------------------|
|              |              |                |                |                              |                       |
|              |              |                |                |                              |                       |

Q.

rpm. & km/hrs.



5 p / rev

5 p /  $2\pi r$   
1 rev

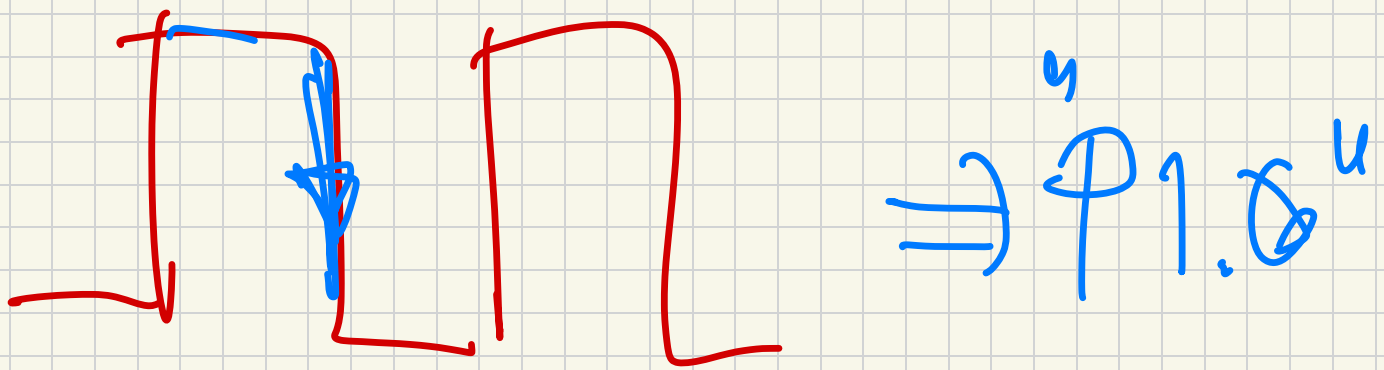
1 p / 0.25m

ESP32

~~X~~ pulse / sec  
A

0.25X km/hr  
km/hr

Max 150 km/hr



Software

loop: JB P1.0, \$

674

INC X

JNB P1.0, \$

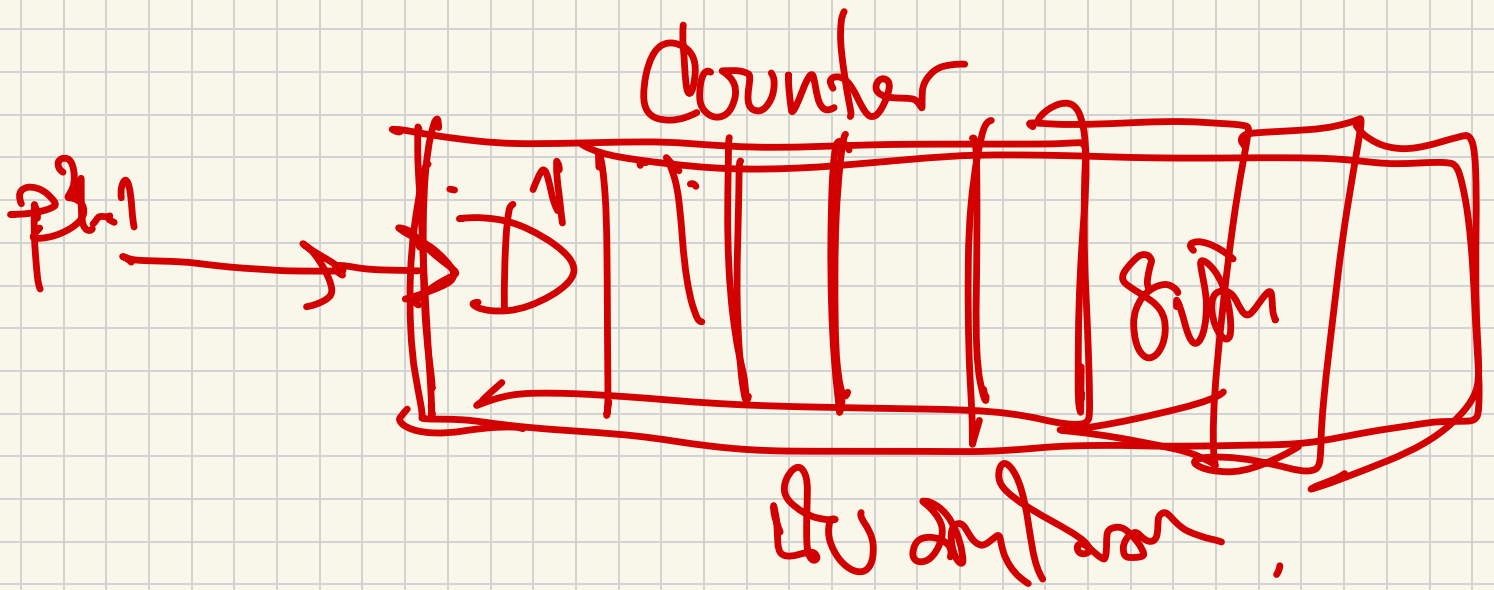
JMP loop

1 MHz

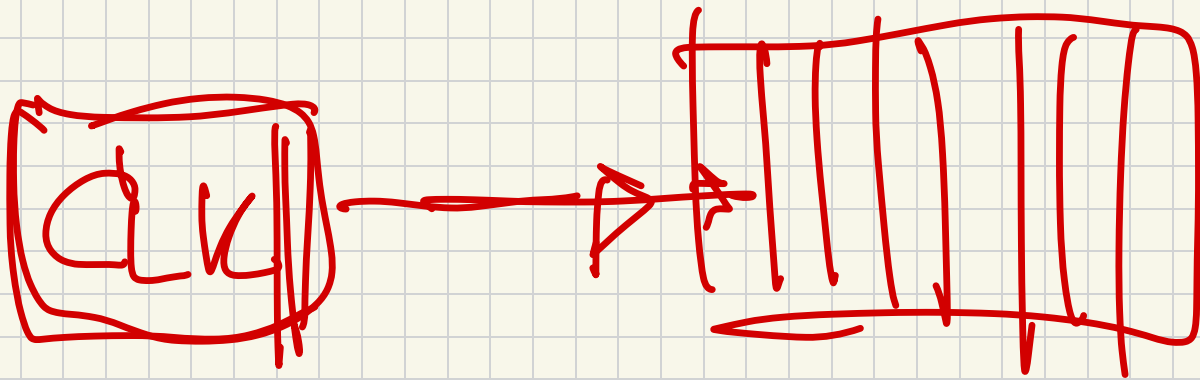
one

X TAL 11.0592 MHz

X TAL = 1 MHz



Counter  $\rightarrow$  D-Flip.



Timer.

1. MCS-51 (A) Timer Delay >> สร้างสัญญาณ จำนวน, เวลาOn, เวลาOff

|                          |
|--------------------------|
| Picture(รูปถ่ายการทำงาน) |
|                          |
| MCS51 Capture Code       |
|                          |
| MCS51 Text Code          |
|                          |

2. ESP-32 (A) Timer Delay >> สร้างสัญญาณ จำนวน, เวลาOn, เวลาOff

|                          |
|--------------------------|
| Picture(รูปถ่ายการทำงาน) |
|                          |
| ESP32 Capture Code       |
|                          |
| ESP32 Text Code          |
|                          |

## 3. MCS-51 (B) Counter with Software &gt;&gt; อ่านสัญญาณ จำนวน, เวลาOn, เวลาOff

|                          |
|--------------------------|
| Picture(รูปถ่ายการทำงาน) |
|                          |
| ESP32 Capture Code       |
|                          |
| ESP32 Text Code          |
|                          |
| MCS51 Capture Code       |
|                          |
| MCS51 Text Code          |
|                          |

## 4. ESP-32 (B) Counter with Software &gt;&gt; อ่านสัญญาณ จำนวน, เวลาOn, เวลาOff

|                          |
|--------------------------|
| Picture(รูปถ่ายการทำงาน) |
|                          |
| ESP32 Capture Code       |
|                          |
| ESP32 Text Code          |
|                          |
| MCS51 Capture Code       |
|                          |
| MCS51 Text Code          |
|                          |

## 5. Node-RED (A)

- จาก Mission-5: From ESP-32 to MQTT Broker and If xxx then LINE(or Telegram) Alert
- หากอุณหภูมิ เกิน xxxx หรือ ความชื้น เกิน yyyy ให้มีการแจ้งเตือนผ่าน LINE

|                             |
|-----------------------------|
| Picture(รูปถ่ายการทำงาน)-1  |
|                             |
| Picture(รูปถ่ายการทำงาน)-2  |
|                             |
| NodeRED: Capture Node       |
|                             |
| NodeRED: Export Flow (Text) |
|                             |
| ESP32 Capture Code          |
|                             |
| ESP32 Text Code             |
|                             |

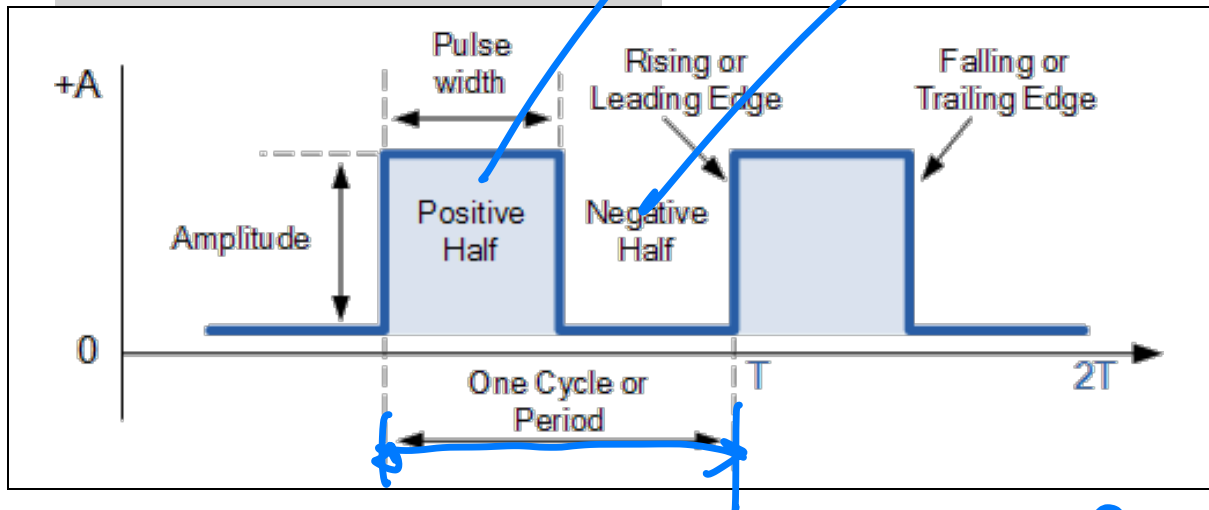
## 6. Node-RED (B)

- จาก Mission-6: From ESP-32 to MQTT Broker and Show on Node-RED Dashboard
- ให้แสดงอุณหภูมิและความชื้นจาก DHT-22 ที่ Node-RED Dashboard
- และสามารถควบคุม LED ได้ 2 ตัว

|                             |
|-----------------------------|
| Picture(รูปถ่ายการทำงาน)-1  |
|                             |
| Picture(รูปถ่ายการทำงาน)-2  |
|                             |
| NodeRED: Dash Board         |
|                             |
| NodeRED: Capture Node       |
|                             |
| NodeRED: Export Flow (Text) |
|                             |
| ESP32 Capture Code          |
|                             |
| ESP32 Text Code             |
|                             |

1/6: เนื้อหา

## 1. Frequency and pulse width on Arduino



- การสังความถี่
  - a. `tone(pin, frequency)`
  - b. `tone(pin, frequency, duration)`
  - c. <https://www.arduitronics.com/article/20/arduino-sing-a-song>
- การอ่านความถี่
  - a. `FreqMeasure.begin();`
  - b. `FreqMeasure.available();`
  - c. `FreqMeasure.read();`
  - d. `FreqMeasure.end();`
  - e. [http://www.ec.in.th/index.php?route=cms/article&article\\_id=36](http://www.ec.in.th/index.php?route=cms/article&article_id=36)

- การสังความกว้าง
  - a. `analogWrite();`
  - b. <https://www.arduino.cc/en/Tutorial/PWM>
  - c. <https://www.arduino.cc/en/Tutorial/SecretsOfArduinoPWM>
  - d. `Servo.attach(pin,min,max)`
  - e. `servo.writeMicroseconds(uS)`
  - f. [www.thaieasyelec.com/article-wiki/review-product-article/บทความตัวอย่างการควบคุม-rc-servo-motor-ด้วย-arduino.html](http://www.thaieasyelec.com/article-wiki/review-product-article/บทความตัวอย่างการควบคุม-rc-servo-motor-ด้วย-arduino.html)
- การอ่านความกว้าง
  - a. `pulseIn();`
  - b. <https://www.arduino.cc/en/Reference/PulseIn>

$T_{on}$   
 $T_{on} + T_{off}$

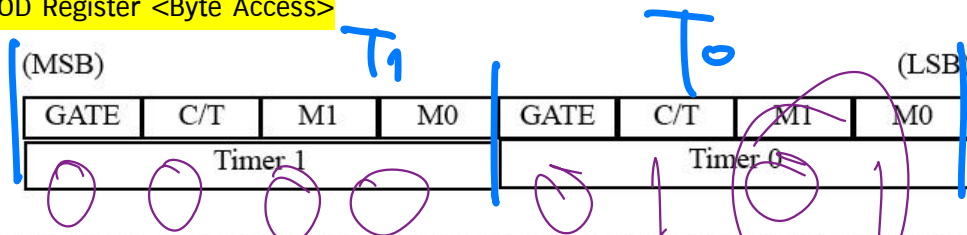
$\times 100$

$$f = \frac{1}{T}$$

→ PWM

## 2. Timer/Counter on MCS-51

### TMOD Register <Byte Access>



**GATE** Gating control when set. The timer/counter is enabled only while the INTx pin is high and the TRx control pin is set. When cleared, the timer is enabled whenever the TRx control bit is set.

**C/T** Timer or counter selected cleared for timer operation (input from internal system clock). Set for counter operation (input from Tx input pin).

**M1** Mode bit 1

**M0** Mode bit 0

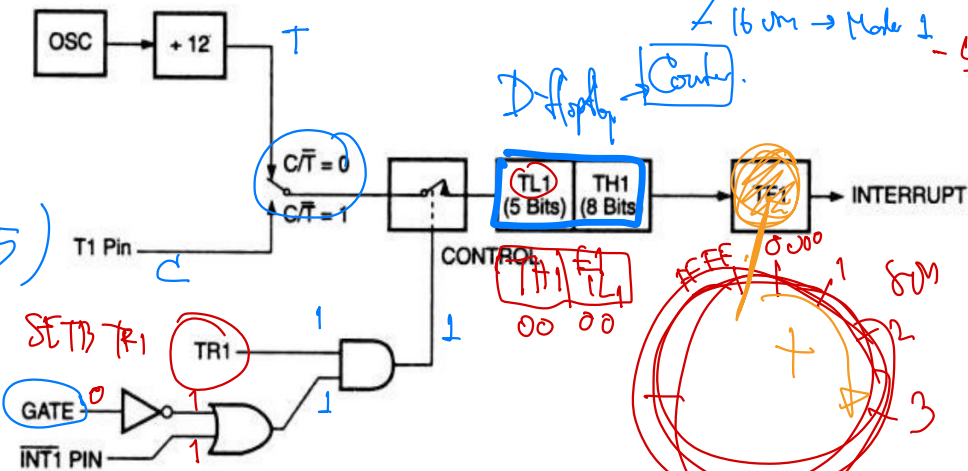
| M1 | M0 | Mode | Operating Mode                                                                                             |
|----|----|------|------------------------------------------------------------------------------------------------------------|
| 0  | 0  | 0    | 13-bit timer mode                                                                                          |
| 0  | 1  | 1    | 8-bit timer/counter THx with TLx as 5-bit prescaler                                                        |
| 0  | 1  | 1    | 16-bit timer mode                                                                                          |
| 1  | 0  | 2    | 16-bit timer/counters THx and TLx are cascaded; there is no prescaler                                      |
| 1  | 0  | 2    | 8-bit auto reload                                                                                          |
| 1  | 0  | 2    | 8-bit auto reload timer/counter; THx holds a value that is to be reloaded into TLx each time it overflows. |
| 1  | 1  | 3    | Split timer mode                                                                                           |

### TCON Register <Bit Access>

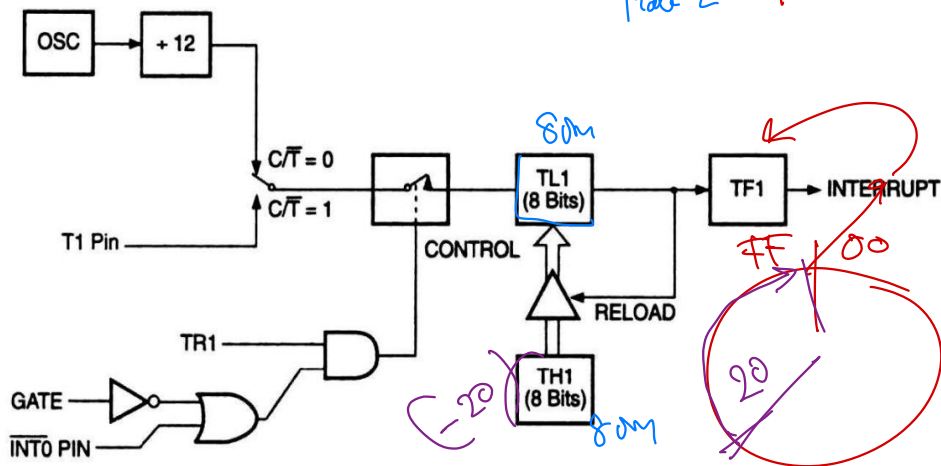
| (MSB) |     |     |     | (LSB) |     |     |     |
|-------|-----|-----|-----|-------|-----|-----|-----|
| TF1   | TR1 | TF0 | TR0 | IE1   | IT1 | IE0 | IT0 |

| Symbol | Position | Name and Significance                                                                                                              | Symbol | Position | Name and Significance                                                                                                  |
|--------|----------|------------------------------------------------------------------------------------------------------------------------------------|--------|----------|------------------------------------------------------------------------------------------------------------------------|
| TF1    | TCON.7   | Timer 1 overflow Flag. Set by hardware on timer/counter overflow. Cleared by hardware when processor vectors to interrupt routine. | IE1    | TCON.3   | Interrupt 1 Edge flag. Set by hardware when external interrupt edge detected. Cleared when interrupt processed.        |
| TR1    | TCON.6   | Timer 1 Run control bit. Set/cleared by software to turn timer/counter on/off.                                                     | IT1    | TCON.2   | Interrupt 1 Type control bit. Set/cleared by software to specify falling edge/low level triggered external interrupts. |
| TF0    | TCON.5   | Timer 0 overflow Flag. Set by hardware on timer/counter overflow. Cleared by hardware when processor vectors to interrupt routine. | IE0    | TCON.1   | Interrupt 0 Edge flag. Set by hardware when external interrupt edge detected. Cleared when interrupt processed.        |
| TR0    | TCON.4   | Timer 0 Run control bit. Set/cleared by software to turn timer/counter on/off.                                                     | IT0    | TCON.0   | Interrupt 0 Type control bit. Set/cleared by software to specify falling edge/low level triggered external interrupts. |

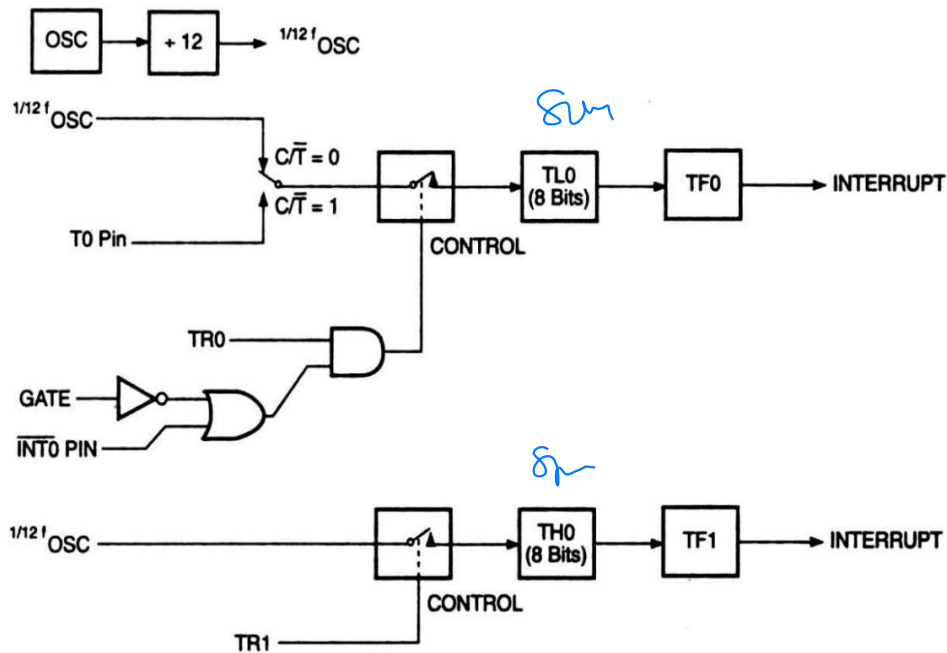
Mode 0: 13Bit , Mode 1: 16 Bit



Mode 2: 8Bit Auto Reload



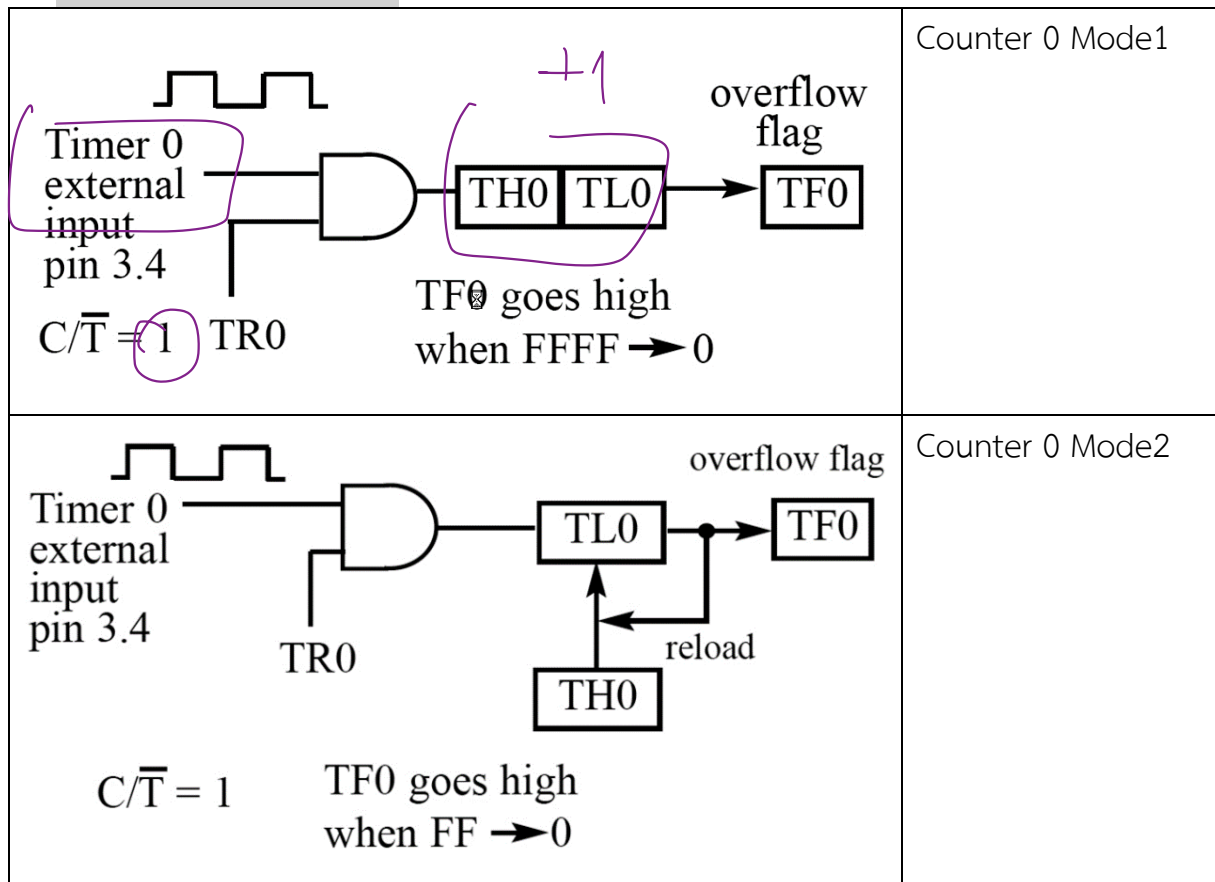
Mode 3: Split 8bit Time and 8Bit Counter



## 3. รีจิสเตอร์ที่สำคัญเกี่ยวกับ Timer/Counter

|              | Bit 7        | Bit 6        | Bit 5       | Bit 4       | Bit 3        | Bit 2      | Bit 1       | Bit 0         |
|--------------|--------------|--------------|-------------|-------------|--------------|------------|-------------|---------------|
| <b>TMOD</b>  | <b>G(T1)</b> | <b>C/T</b>   | <b>M1</b>   | <b>M0</b>   | <b>G(T0)</b> | <b>C/T</b> | <b>M1</b>   | <b>M0</b>     |
| <b>TCON</b>  | <b>TF1</b>   | <b>TR1</b>   | <b>TF0</b>  | <b>TR0</b>  | <b>IE1</b>   | <b>IT1</b> | <b>IE0</b>  | <b>IT0</b>    |
| <b>SCON</b>  | <b>SM0</b>   | <b>SM1</b>   | <b>SM2</b>  | <b>REN</b>  | <b>TB8</b>   | <b>RB8</b> | <b>TI</b>   | <b>RI</b>     |
| <b>PCON</b>  | <b>SMOD1</b> | <b>SMOD0</b> | -           | <b>POF</b>  | <b>GF1</b>   | <b>GF0</b> | <b>PD</b>   | <b>IDL</b>    |
| <b>T2CON</b> | <b>TF2</b>   | <b>EXF2</b>  | <b>RCLK</b> | <b>TCLK</b> | <b>EXEN2</b> | <b>TR2</b> | <b>C/T2</b> | <b>CP/RL2</b> |

## 4. Counter Configuration



ตัวอย่าง การใช้ในโหมด Counter

```

MOV  TMOD,#00000101B    ; Using Counter Mode 1 16 Bit
MOV  TH0,#0              ; Initial Value = 0000
MOV  TL0,#0
SETB TR0                 ; Start Counter

Loop: {Show TH0TL0}      ; Loop for Show Value of TH0TL0
      JMP  Loop

```

โดย ดูจาก 107

TH0 TL0 255

## 5. Timer Mode1 Configuration

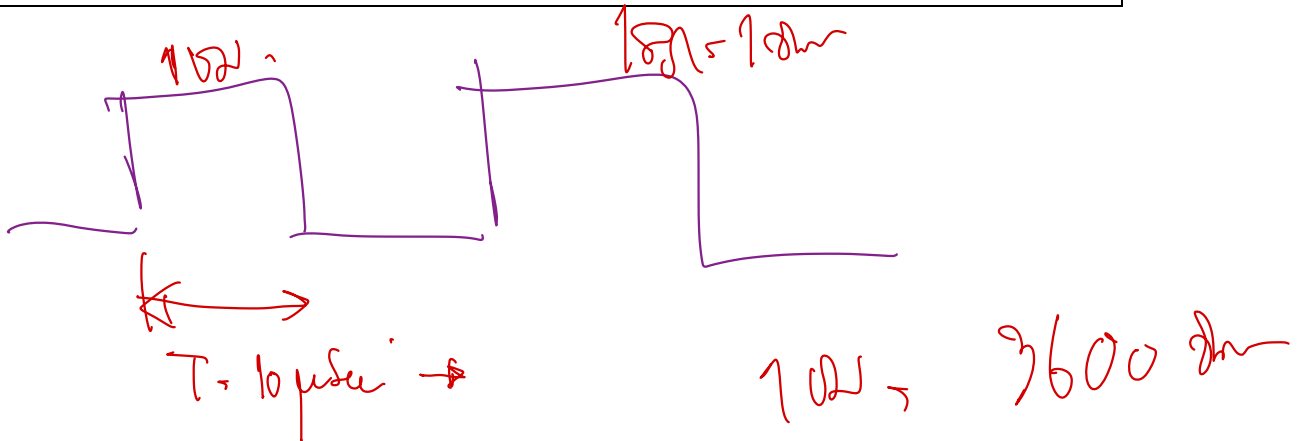
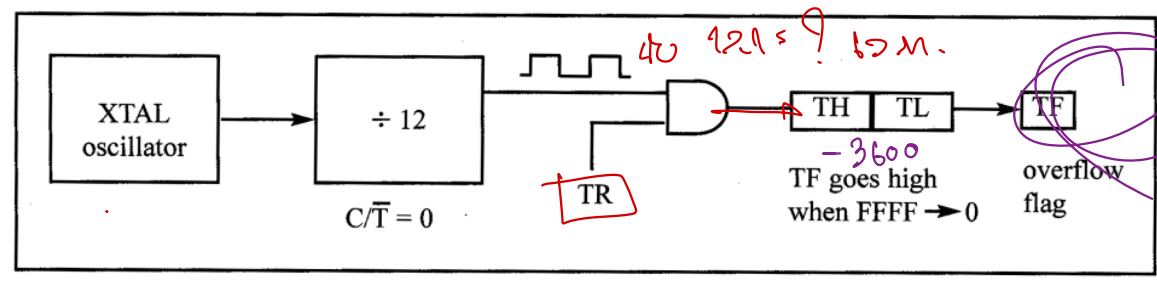
## Steps to program in mode 1

1. Load the TMOD value
2. Load registers TL and TH
3. Start the timer (SETB TR0 or SETB TR1)  
**< Work >**
4. Keep monitoring the timer flag (TF) with the "JNB TFx,\$"
5. Stop the timer (CLR TR0 or CLR TR1)
6. Clear the TF flag
7. Go back to step 2

```

MOV    TMOD,#00000001B
LOOP:  MOV    TH0,#0F2H
        MOV    TLO,#0FFH
        SETB   TR0
        CPL    P1
        JNB    TFO,$

        CLR    TR0
        CLR    TFO
        JMP    LOOP
  
```



## 6. Timer Mode2 Configuration

| Steps to program in mode 2                                                                                                                                                     |                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1. Load the TMOD value<br>2. Load the TH registers<br>3. Start the timer<br>4. Keep monitoring the timer flag (TF)<br>< Work ><br>5. Clear the TF flag<br>6. Go back to step 4 | <pre> MOV    TMOD,#20H MOV    TH1,#05H SETB   TR1 ✓ LOOP:  JNB   TF0,\$         CPL   P1         CLR   TF0         JMP   LOOP </pre> |
|                                                                                                                                                                                |                                                                                                                                      |

Dim (1) → 0000 → FFFF

65535 d1

Dim (2) 00 → FF

256 d1

### ตัวอย่าง.1 การใช้ Timer ในการหน่วงเวลา

เช่น ต้องการสร้างความถี่ 10 kHz  $\rightarrow$  Period = 0.1mSec  $\rightarrow$  Period/2 = 50uSec  
 Delay 50uSec ที่บอร์ด Crystal 11.0592MHz

$$\rightarrow 1MC = 12/11.0592 = 1.085uSec$$

$$\rightarrow 50uSec = 46.08 MC$$

MOV DPTR, #-42

MOV TH0, DPTR

MOV TL0, DPTR

**; Load Timer0 = 0-46 = -46 = FFD2H**

**MOV TMOD, #01 ; Put Timer 0 in 16-bit mode**

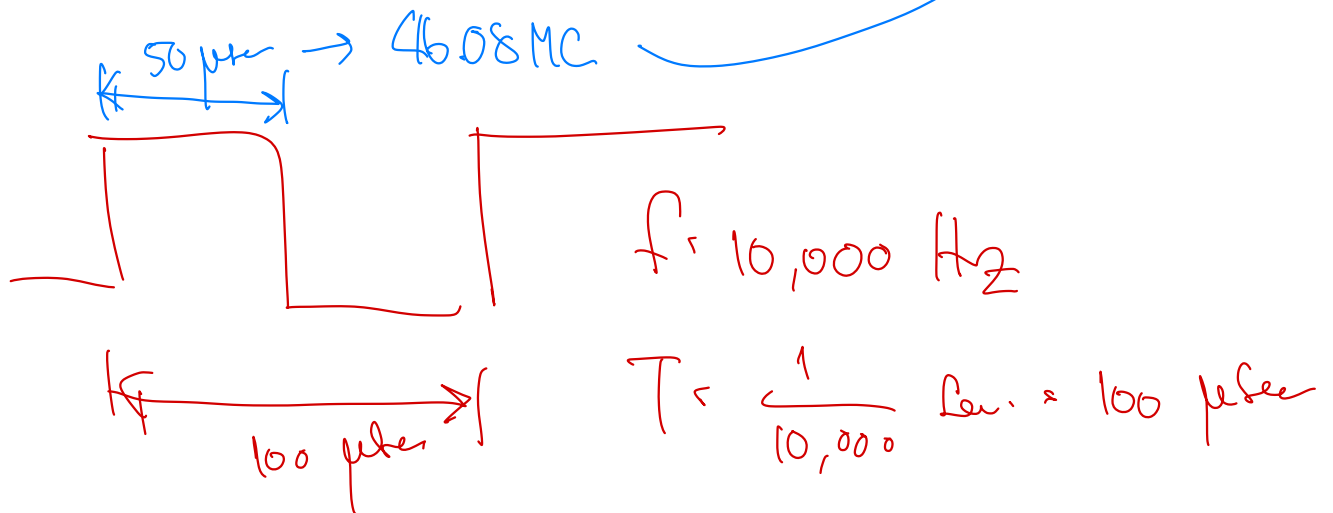
**MOV TH0, #0FFH ; High byte of -46(FFD2H)**

**MOV TL0, #0D2H ; Low byte of -46(FFD2H)**

**SETB TR0 ; Make Timer 0 Start Counting**

**CLR TF0**

**JNB TF0, \$ ; If TF0 is not set, jump back to this same ; instruction**



## ตัวอย่าง.2 การใช้ Timer ในการหน่วงเวลาสำหรับ Serial Communication

### Using Timer1, Mode2

<http://www.intel.com/design/mcs51/applnots/270490.htm>

Serial Mode1,3 และใช้ Timer 1 Mode2 →

$$\text{BaudRate} = \frac{2^{\text{SMOD1}} \times \text{Oscillator Frequency}}{32 \times 12 \times [256 - (\text{th1})]}$$

9600 @18.432MHz → SMOD1=0, TH1=251(FBH)

9600 @11.0592MHz → SMOD1=0, TH1=253(FDH)

```
MOV SCON,#050H      ; Serial Port Mode 3
ANL PCON,#7FH        ; SMOD 1=2
MOV TMOD,#20H        ; Timer1 Mode2
MOV TH1,#0FDH        ; Reload value for desired baud
SETB TR1             ; Turn on Timer1
```

Serial Baud (115200);

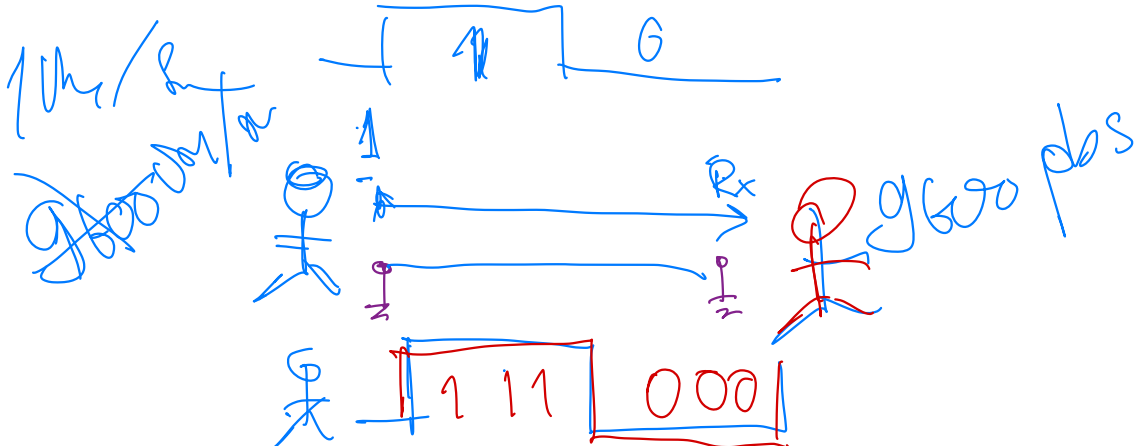
### Using Timer2

$$\text{Baud rate} = \frac{\text{Oscillator Frequency}}{32 \times [65536 - (\text{RCAP2H}, \text{RCAP2L})]}$$

9600 @18.432MHz → RCAP2H, RCAP2L = 65476 = FFC4H

9600 @11.0592MHz → RCAP2H, RCAP2L = 65500 = FFDCH

```
MOV SCON,#01010000B ; Mode1 Serial Port
MOV RCAP2H,#0FFH    ; Reload values for desired
MOV RCAP2L,#0DCH    ; baud rate 9600@11.0592
MOV T2CON,#00110100B ; Timer 2 as baud rate generator
                     ; and turn on Timer 2
```



○ 0° 30H  
人

人

110000

00110000 00110001 00110010

30  
'0'

31  
'1'

32  
'1'

00110000 00110001 00110010

ตัวอย่าง.3 Timer.C <Ride-51> polling Delay

```

//*****
// XTAL : 18.4322 MHz
// 1 MC = 12/18.432MHz = 0.651041667 uSec
// 1000 uSec = 1 mSec = 1536 MC --> 1536
// 65530-1536 = 64000 = 0xFA00
//*****

#include <REG51.H>
void dmsec(unsigned char time)
{
    unsigned char i;
    for (i=0;i<time;i++)
    {
        TH0 = 0xFA; // +18 For move & compare instruction
        TL0 = 0x00+18;
        TF0 = 0;
        TR0 = 1;
        while(TF0 == 0);
        TR0 = 0;
    }
}

void main()
{
    TMOD = 0x01;
    while(1)
    {
        P2 ^= 0x0ff;
        dmsec(50);
    }
}

```

$-600+2$   
 $-1536$  MC  
 $me$   
 $10\mu s \Rightarrow 600$

ตัวอย่าง.4 *Timer.C* <Ride-51> interrupts Delay

```

//*****//
// XTAL : 18.4322 MHz
// 1 MC = 12/18.432MHz = 0.651041667 uSec
// 1000 uSec = 1 mSec = 1536 MC --> 1536
// 65530-1536 = 64000 = 0xFA00
//*****//

#include <REG51.H>
int const_Delay,count_Delay;
void Timer0_isr(void) interrupt 1 {
    TR0 = 0;
    TH0 = 0xFA;
    TL0 = 0x00+18;    // +18 For move & compare instruction
    TR0 = 1;
    count_Delay++;
    if (count_Delay>=const_Delay) {
        P2 ^= 0x0ff;
        count_Delay = 0;
    }
}

void main() {
    //***** Main
    const_Delay = 50; // Delay 50mS = 1/(2x50mS) = 10.00Hz
    TMOD = 0x01;    //Timer 16 bit
    ET0 = 1;
    TR0 = 1;
    EA = 1;
    while(1) {
    }
}

```

ตัวอย่าง.5 Timer.C <Ride-51> timer0 mode2 interrupts for scan display

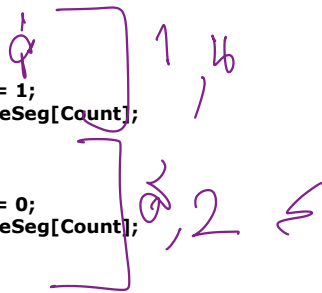
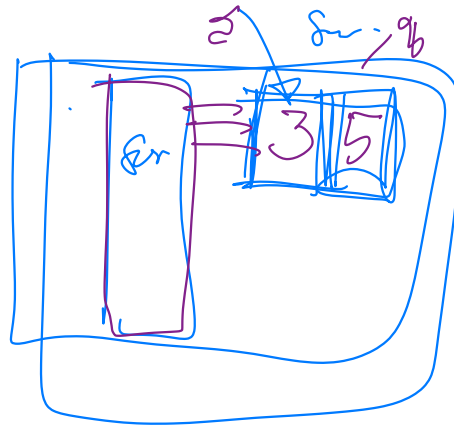
```
#include <REG51.H>
#define PortO_DispData P0
sbit CS_Seg0 = P1 ^ 4;
sbit CS_Seg1 = P1 ^ 5;
code unsigned char CodeSeg[] =
{ 0x77, 0x11, 0x6D, 0x5D, 0x1B, 0x5E, 0x7E, 0x15,
  0x7F, 0x5F, 0x3F, 0x7A, 0x66, 0x79, 0x6E, 0x2E
};
```

```
unsigned int timerCount, Counter = 0;
```

```
void Delay(unsigned char msec)
{ unsigned char i, j;
  while (msec)
  { for (i = 1; i <= 250; i++) //not :
    for (j = 1; j <= 113; j++); //with ;
    msec--;
  }
}
```

```
void Timer0_isr(void) interrupt 1 {
  unsigned char Count;
  timerCount++;
  timerCount = timerCount % 2;
  if(timerCount==0){
    Count = Counter/10;
    CS_Seg0 = 0; CS_Seg1 = 1;
    PortO_DispData = ~CodeSeg[Count];
  }
  if(timerCount==1){
    Count = Counter%10;
    CS_Seg0 = 1; CS_Seg1 = 0;
    PortO_DispData = ~CodeSeg[Count];
  }
}
```

```
void main() {
  TMOD = 0x02;
  TH0 = 0x00;
  TL0 = 0x00;
  ET0 = 1;
  TR0 = 1;
  EA = 1;
  while(1) {
    Counter++;
    if(Counter>99) Counter=0;
    Delay(25);
  }
}
```



// Timer0 Mode2

## ตัวอย่าง.6 Lab8\_Timer.Asm

สร้างเวลาที่ 1 kHz = 1000 MicroSec

Duty Cycle 50% Ton=Toff = 500 MicroSec

=  $500 * 18.432 / 12$  MC 12 Clock/MC

= 768 MC

=  $500 * 18.432 / 6$  MC x2 Mode = 6 Clock/MC

= 1536 MC

| ;==== Lab8_Timer.Asm ===== |      |                 |                    |    |
|----------------------------|------|-----------------|--------------------|----|
|                            | ORG  | 0000H           |                    |    |
|                            | JMP  | 0100H           |                    |    |
|                            | ORG  | 0100H           |                    |    |
|                            | MOV  | SP,#3FH         |                    |    |
|                            | CLR  | EA              |                    |    |
|                            | MOV  | TMOD,#00010001B |                    |    |
| T_OFF:                     | CLR  | TR1             | ; Stop Timer       | 4  |
|                            | MOV  | A,#00H          | ; Out Pulse        | 5  |
|                            | MOV  | P2,A            | ; Out P1           | 6  |
|                            | MOV  | DPTR,#-1536+13  | ; Reload Data      | 8  |
|                            | MOV  | TH1,DPH         | ; For TH1          | 9  |
|                            | MOV  | TL1,DPL         | ; For TL1          | 10 |
|                            | CLR  | TF1             | ; Initial Start    | 11 |
|                            | SETB | TR1             | ; Start Timer1     | 12 |
|                            | JNB  | TF1,\$          | ; Wait Until TF On | 13 |
|                            |      |                 |                    | 14 |
| T_ON:                      | CLR  | TR1             | ; 1MC              | 15 |
|                            | MOV  | A,#0FFH         | ; 1MC              | 16 |
|                            | MOV  | P2,A            | ; 1MC              | 17 |
|                            | MOV  | DPTR,#-1536+15  | ; 2MC              | 20 |
|                            | MOV  | TH1,DPH         | ; 2MC              | 21 |
|                            | MOV  | TL1,DPL         | ; 2MC              | 22 |
|                            | CLR  | TF1             | ; 1MC              | 23 |
|                            | SETB | TR1             | ; 1MC              | 24 |
|                            | JNB  | TF1,\$          | ; 2MC              | 25 |
|                            |      |                 |                    | 26 |
|                            | JMP  | T_OFF           | ; 2MC              | 27 |
|                            | END  |                 |                    |    |

## ตัวอย่าง.7 Lab8\_Counter1.Asm

โปรแกรมสำหรับอ่านค่าการนับสัญญาณพัลส์ที่ P3.4 โดยไม่ใช้เทคนิค Counter

```

;==== Lab8_Counter1.Asm ====

Port_7Seg      EQU      P0
P_SegLeft      EQU      P1.4
P_SegRight     EQU      P1.5
P_TestIn       EQU      P3.4
Counter        EQU      20H

                ORG      0000H
                JMP      0100H

                ORG      0100H
                MOV      SP,#3FH
                CLR      EA
                SETB     P_SegLeft
                CLR      P_SegRight
                SETB     P_TestIn
                MOV      Counter,#0

Main_Loop:      CALL     Display
                JB       P_TestIn,$
                JNB      P_TestIn,$
                INC      Counter
                JMP      Main_Loop

Display:        MOV      A,Counter
                ANL      A,#0FH
                MOV      DPTR,#CODE_TABLE
                MOVC     A,@A+DPTR
                CPL      A
                MOV      Port_7Seg,A
                RET

CODE_TABLE:    DB       77H, 11H, 6DH, 5DH
                DB       1BH, 5EH, 7EH, 15H
                DB       7FH, 5FH, 3FH, 7AH
                DB       66H, 79H, 6EH, 2EH

Delay_XXX:     CLR      A
                MOV      B,#200
_Dly00:        DJNZ     Acc,_Dly00
                DJNZ     B,_Dly00
                RET

                END

```

**Diagram:** A timing diagram for the P3.4 input port. It shows a square wave pulse. A purple arrow points from the 'Main\_Loop' section of the code to the rising edge of the pulse, labeled '1'. Another purple arrow points from the 'Main\_Loop' section to the falling edge of the pulse, labeled '0'. A blue arrow points from the 'Display' section to the pulse, labeled '2'. A blue circle with a plus sign and '1' is shown next to the pulse, indicating an increment operation.

## ตัวอย่าง.8 Lab8\_Counter2.Asm

โปรแกรมสำหรับอ่านค่าการนับสัญญาณพัลส์ที่ P3.4 (T0) โดยใช้เทคนิค Counter

|                                   |             |                           |
|-----------------------------------|-------------|---------------------------|
| <b>;=== Lab8_Counter2.Asm ===</b> |             |                           |
| <b>Port_7Seg</b>                  | <b>EQU</b>  | <b>P0</b>                 |
| <b>P_SegLeft</b>                  | <b>EQU</b>  | <b>P1.4</b>               |
| <b>P_SegRight</b>                 | <b>EQU</b>  | <b>P1.5</b>               |
| <b>P_TestIn</b>                   | <b>EQU</b>  | <b>P3.4</b>               |
| <b>Counter</b>                    | <b>EQU</b>  | <b>20H</b>                |
|                                   | <b>ORG</b>  | <b>0000H</b>              |
|                                   | <b>JMP</b>  | <b>0100H</b>              |
|                                   | <b>ORG</b>  | <b>0100H</b>              |
|                                   | <b>MOV</b>  | <b>SP,#3FH</b>            |
|                                   | <b>CLR</b>  | <b>EA</b>                 |
|                                   | <b>SETB</b> | <b>P_SegLeft</b>          |
|                                   | <b>CLR</b>  | <b>P_SegRight</b>         |
|                                   | <b>SETB</b> | <b>P_TestIn</b>           |
|                                   | <b>MOV</b>  | <b>TMOD,#00000101B</b>    |
|                                   | <b>MOV</b>  | <b>TH0,#0</b>             |
|                                   | <b>MOV</b>  | <b>TL0,#0</b>             |
|                                   | <b>SETB</b> | <b>TR0</b>                |
| <b>Main_Loop:</b>                 | <b>MOV</b>  | <b>Counter,TL0</b>        |
|                                   | <b>CALL</b> | <b>Display</b>            |
|                                   | <b>JMP</b>  | <b>Main_Loop</b>          |
| <b>Display:</b>                   | <b>MOV</b>  | <b>A,Counter</b>          |
|                                   | <b>ANL</b>  | <b>A,#0FH</b>             |
|                                   | <b>MOV</b>  | <b>DPTR,#CODE_TABLE</b>   |
|                                   | <b>MOVC</b> | <b>A,@A+DPTR</b>          |
|                                   | <b>CPL</b>  | <b>A</b>                  |
|                                   | <b>MOV</b>  | <b>Port_7Seg,A</b>        |
|                                   | <b>RET</b>  |                           |
| <b>CODE_TABLE:</b>                | <b>DB</b>   | <b>77H, 11H, 6DH, 5DH</b> |
|                                   | <b>DB</b>   | <b>1BH, 5EH, 7EH, 15H</b> |
|                                   | <b>DB</b>   | <b>7FH, 5FH, 3FH, 7AH</b> |
|                                   | <b>DB</b>   | <b>66H, 79H, 6EH, 2EH</b> |
| <b>Delay_XXX:</b>                 | <b>CLR</b>  | <b>A</b>                  |
|                                   | <b>MOV</b>  | <b>B,#200</b>             |
| <b>_Dly00:</b>                    | <b>DJNZ</b> | <b>Acc,_Dly00</b>         |
|                                   | <b>DJNZ</b> | <b>B,_Dly00</b>           |
|                                   | <b>RET</b>  |                           |
|                                   | <b>END</b>  |                           |



## ตัวอย่าง.9 Lab8\_Counter3.Asm

โปรแกรมสำหรับอ่านค่าการนับสัญญาณพัลส์ที่ P3.4 (T0) โดยใช้เทคนิค Counter

| ;== Lab8_Counter3.Asm ,Long== |      |                       | ;== Lab8_Counter3.Asm ,Short == |      |                       |
|-------------------------------|------|-----------------------|---------------------------------|------|-----------------------|
| Port_7Seg                     | EQU  | P0                    | Port_7Seg                       | EQU  | P0                    |
| P_SegLeft                     | EQU  | P1.4                  | P_SegLeft                       | EQU  | P1.4                  |
| P_SegRight                    | EQU  | P1.5                  | P_SegRight                      | EQU  | P1.5                  |
| P_TestIn                      | EQU  | P3.4                  | P_TestIn                        | EQU  | P3.4                  |
| Counter                       | EQU  | 20H                   | Counter                         | EQU  | 20H                   |
|                               | ORG  | 0000H                 |                                 | ORG  | 0000H                 |
|                               | JMP  | 0100H                 |                                 | JMP  | 0100H                 |
|                               | ORG  | 0100H                 |                                 | ORG  | 0100H                 |
|                               | MOV  | SP,#3FH               |                                 | MOV  | SP,#3FH               |
|                               | CLR  | EA                    |                                 | CLR  | EA                    |
|                               | SETB | P_TestIn ; Input Port |                                 | SETB | P_TestIn ; Input Port |
|                               | MOV  | TMOD,#00000101B       |                                 | MOV  | TMOD,#00000101B       |
|                               | MOV  | TH0,#0                |                                 | MOV  | TH0,#0                |
|                               | MOV  | TL0,#0                |                                 | MOV  | TL0,#0                |
|                               | SETB | TR0                   |                                 | SETB | TR0                   |
| Main_Loop:                    | MOV  | Counter,TL0           | Main_Loop:                      | MOV  | Counter,TL0           |
|                               | CALL | Display               |                                 | CALL | Display               |
|                               | JMP  | Main_Loop             |                                 | JMP  | Main_Loop             |
| Display:                      | MOV  | A,Counter             | Display:                        | MOV  | A,Counter             |
|                               | SETB | P_SegLeft             |                                 | SETB | P_SegLeft             |
|                               | CLR  | P_SegRight            |                                 | CLR  | P_SegRight            |
|                               | CALL | _DISP00               |                                 | CALL | _DISP00               |
|                               | MOV  | A,Counter             |                                 | MOV  | A,Counter             |
|                               | SWAP | A                     |                                 | SWAP | A                     |
|                               | CLR  | P_SegLeft             |                                 | CLR  | P_SegLeft             |
|                               | SETB | P_SegRight            |                                 | SETB | P_SegRight            |
|                               | CALL | _DISP00               | _DISP00:                        | ANL  | A,#0FH                |
|                               | RET  |                       |                                 | MOV  | DPTR,#CODE_TABLE      |
| _DISP00:                      | ANL  | A,#0FH                |                                 | MOVC | A,@A+DPTR             |
|                               | MOV  | DPTR,#CODE_TABLE      |                                 | CPL  | A                     |
|                               | MOVC | A,@A+DPTR             |                                 | MOV  | Port_7Seg,A           |
|                               | CPL  | A                     |                                 | CLR  | A                     |
|                               | MOV  | Port_7Seg,A           |                                 | DJNZ | Acc,\$                |
|                               | CLR  | A                     |                                 | RET  |                       |
|                               | DJNZ | Acc,\$                |                                 |      |                       |
|                               | RET  |                       |                                 |      |                       |
| CODE_TABLE:                   | DB   | 77H, 11H, 6DH, 5DH    | CODE_TABLE:                     | DB   | 77H, 11H, 6DH, 5DH    |
|                               | DB   | 1BH, 5EH, 7EH, 15H    |                                 | DB   | 1BH, 5EH, 7EH, 15H    |
|                               | DB   | 7FH, 5FH, 3FH, 7AH    |                                 | DB   | 7FH, 5FH, 3FH, 7AH    |
|                               | DB   | 66H, 79H, 6EH, 2EH    |                                 | DB   | 66H, 79H, 6EH, 2EH    |
| Delay_XXX:                    | CLR  | A                     | Delay_XXX:                      | CLR  | A                     |
|                               | MOV  | B,#200                |                                 | MOV  | B,#200                |
| _Dly00:                       | DJNZ | Acc,_Dly00            | _Dly00:                         | DJNZ | Acc,_Dly00            |
|                               | DJNZ | B,_Dly00              |                                 | DJNZ | B,_Dly00              |
|                               | RET  |                       |                                 | RET  |                       |
|                               | END  |                       |                                 | END  |                       |

## ตัวอย่าง.10 Timer Interrupts

โปรแกรมสแกนหน้าจอ 7Segment ด้วยการใช้ Timer Interrupts

|                       |     |       |
|-----------------------|-----|-------|
| Port_7Seg             | EQU | P0    |
| P_SegLeft             | EQU | P1.4  |
| P_SegRight            | EQU | P1.5  |
| Counter               | EQU | 20H   |
| BitDispLR             | EQU | 21H.0 |
| ORG 0000H             |     |       |
| JMP 0100H             |     |       |
| ORG 000BH             |     |       |
| JMP TO_ISR            |     |       |
| ORG 0100H             |     |       |
| CLR EA                |     |       |
| MOV SP,#3FH           |     |       |
| MOV Counter,#00H      |     |       |
| MOV TMOD,#02H         |     |       |
| MOV TH0,#0            |     |       |
| MOV TL0,#0            |     |       |
| SETB ET0              |     |       |
| SETB EA               |     |       |
| SETB TR0              |     |       |
| Main_Loop:            |     |       |
| INC Counter           |     |       |
| CALL Delay            |     |       |
| JMP Main_Loop         |     |       |
| Delay:                |     |       |
| MOV R7,#20            |     |       |
| MOV A,#00             |     |       |
| MOV B,#00             |     |       |
| _Dly00:               |     |       |
| DJNZ B,_Dly00         |     |       |
| DJNZ Acc,_Dly00       |     |       |
| DJNZ R7,_Dly00        |     |       |
| RET                   |     |       |
| TO_ISR:               |     |       |
| PUSH Acc              |     |       |
| PUSH DPH              |     |       |
| PUSH DPL              |     |       |
| MOV A,Counter         |     |       |
| JB BitDispLR,_DRight  |     |       |
| _DLeft:               |     |       |
| SETB P_SegLeft        |     |       |
| CLR P_SegRight        |     |       |
| JMP _DShow            |     |       |
| _DRight:              |     |       |
| CLR P_SegLeft         |     |       |
| SETB P_SegRight       |     |       |
| SWAP A                |     |       |
| _DShow:               |     |       |
| ANL A,#0FH            |     |       |
| MOV DPTR,#CODE_TABLE  |     |       |
| MOVC A,@A+DPTR        |     |       |
| CPL A                 |     |       |
| MOV Port_7Seg,A       |     |       |
| CPL BitDispLR         |     |       |
| POP DPL               |     |       |
| POP DPH               |     |       |
| POP Acc               |     |       |
| RETI                  |     |       |
| CODE_TABLE:           |     |       |
| DB 77H, 11H, 6DH, 5DH |     |       |
| DB 1BH, 5EH, 7EH, 15H |     |       |
| DB 7FH, 5FH, 3FH, 7AH |     |       |
| DB 66H, 79H, 6EH, 2EH |     |       |
| END                   |     |       |

ตัวอย่าง.11 Timer2 Interrupt < Ref=<http://web.mit.edu/6.115/www/document/8051.pdf> >

### T2CON: Timer/Counter 2 Control Register. Bit Addressable 8052 Only

| TF2 | EXF2 | RCLK | TCLK | EXEN2 | TR2 | C/T2 | CP/RL2 |
|-----|------|------|------|-------|-----|------|--------|
|-----|------|------|------|-------|-----|------|--------|

|        |          |                                                                                                                                                                                                                                                                                                                                        |  |  |  |  |  |
|--------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| TF2    | T2CON. 7 | Timer 2 overflow flag set by hardware and cleared by software. TF2 cannot be set when either RCLK = 1 or CLK = 1                                                                                                                                                                                                                       |  |  |  |  |  |
| EXF2   | T2CON. 6 | Timer 2 external flag set when either a capture or reload is caused by a negative transition on T2EX, and EXEN2 = 1. When Timer 2 interrupt is enabled, EXF2 = 1 will cause the CPU to vector to the Timer 2 interrupt routine. EXF2 must be cleared by software.                                                                      |  |  |  |  |  |
| RCLK   | T2CON. 5 | Receive clock flag. When set, causes the Serial Port to use Timer 2 overflow pulses for its receive clock in modes 1 & 3. RCLK = 0 causes Timer 1 overflow to be used for the receive clock.                                                                                                                                           |  |  |  |  |  |
| TCLK   | T2CON. 4 | Transmit clock flag. When set, causes the Serial Port to use Timer 2 overflow pulses for its transmit clock in modes 1 & 3. TCLK = 0 causes Timer 1 overflows to be used for the transmit clock.                                                                                                                                       |  |  |  |  |  |
| EXEN2  | T2CON. 3 | Timer 2 external enable flag. When set, allows a capture or reload to occur as a result of negative transition on T2EX if Timer 2 is not being used to clock the Serial Port. EXEN2 = 0 causes Timer 2 to ignore events at T2EX.                                                                                                       |  |  |  |  |  |
| TR2    | T2CON. 2 | Software START/STOP control for Timer 2. A logic 1 starts the Timer.                                                                                                                                                                                                                                                                   |  |  |  |  |  |
| C/T2   | T2CON. 1 | Timer or Counter select.<br>0 = Internal Timer. 1 = External Event Counter (falling edge triggered).                                                                                                                                                                                                                                   |  |  |  |  |  |
| CP/RL2 | T2CON. 0 | Capture/Reload flag. When set, captures will occur on negative transitions at T2EX if EXEN2 = 1. When cleared, Auto-Reloads will occur either with Timer 2 overflows or negative transitions at T2EX when EXEN2 = 1. When either RCLK = 1 or TCLK = 1, this bit is ignored and the Timer is forced to Auto-Reload on Timer 2 overflow. |  |  |  |  |  |

### Timer/Counter 2 Setup

Except for the baud rate generator mode, the values given for T2CON do not include the setting of the TR2 bit. Therefore, bit TR2 must be set, separately, to turn the Timer on.

#### As a Timer:

Table 7

| MODE                                                  | T2CON                     |                           |
|-------------------------------------------------------|---------------------------|---------------------------|
|                                                       | INTERNAL CONTROL (NOTE 1) | EXTERNAL CONTROL (NOTE 2) |
| 16-bit Auto-Reload                                    | 00H                       | 08H                       |
| 16-bit Capture                                        | 01H                       | 09H                       |
| BAUD rate generator receive & transmit same baud rate | 34H                       | 36H                       |
| receive only                                          | 24H                       | 26H                       |
| transmit only                                         | 14H                       | 16H                       |

#### As a Counter:

Table 8

| MODE               | TMOD                      |                           |
|--------------------|---------------------------|---------------------------|
|                    | INTERNAL CONTROL (NOTE 1) | EXTERNAL CONTROL (NOTE 2) |
| 16-bit Auto-Reload | 02H                       | 0AH                       |
| 16-bit Capture     | 03H                       | 0BH                       |

#### NOTES:

1. Capture/Reload occurs only on Timer/Counter overflow.
2. Capture/Reload occurs on Timer/Counter overflow and a 1 to 0 transition on T2EX (P1.1) pin except when Timer 2 is used in the baud rate generating mode.

**8052 Register Byte Address**

|    |       |      |        |        |     |     |  |      |    |
|----|-------|------|--------|--------|-----|-----|--|------|----|
| 80 | P0    | SP   | DPL    | DPH    |     |     |  | PCON | 87 |
| 88 | TCON  | TMOD | TL0    | TL1    | TH0 | TH1 |  |      | 8F |
| 90 | P1    |      |        |        |     |     |  |      | 97 |
| 98 | SCON  | SBUF |        |        |     |     |  |      | 9F |
| A0 | P2    |      |        |        |     |     |  |      | A7 |
| A8 | IE    |      |        |        |     |     |  |      | AF |
| B0 | P3    |      |        |        |     |     |  |      | B7 |
| B8 | IP    |      |        |        |     |     |  |      | B9 |
| C0 |       |      |        |        |     |     |  |      | C7 |
| C8 | T2CON |      | RCAP2L | RCAP2H | TL2 | TH2 |  |      | CF |
| D0 | PSW   |      |        |        |     |     |  |      | D7 |
| D8 |       |      |        |        |     |     |  |      | DF |
| E0 | ACC   |      |        |        |     |     |  |      | E7 |
| E8 |       |      |        |        |     |     |  |      | EF |
| F0 | R     |      |        |        |     |     |  |      | F7 |
| F8 |       |      |        |        |     |     |  |      | FF |



Blue background are I/O port SFRs  
Yellow background are control SFRs  
Green background are other SFRs

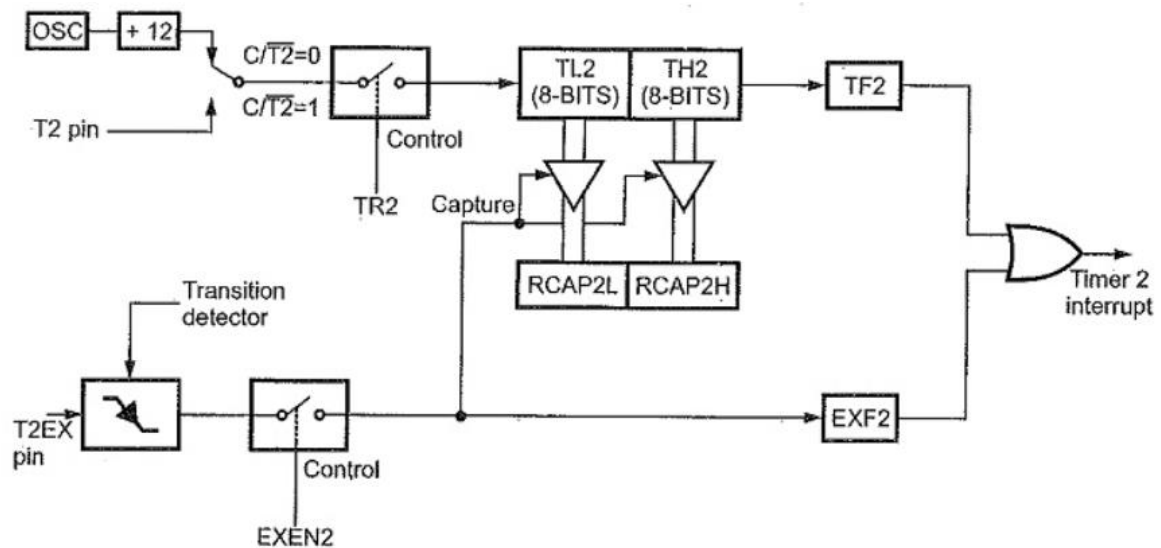
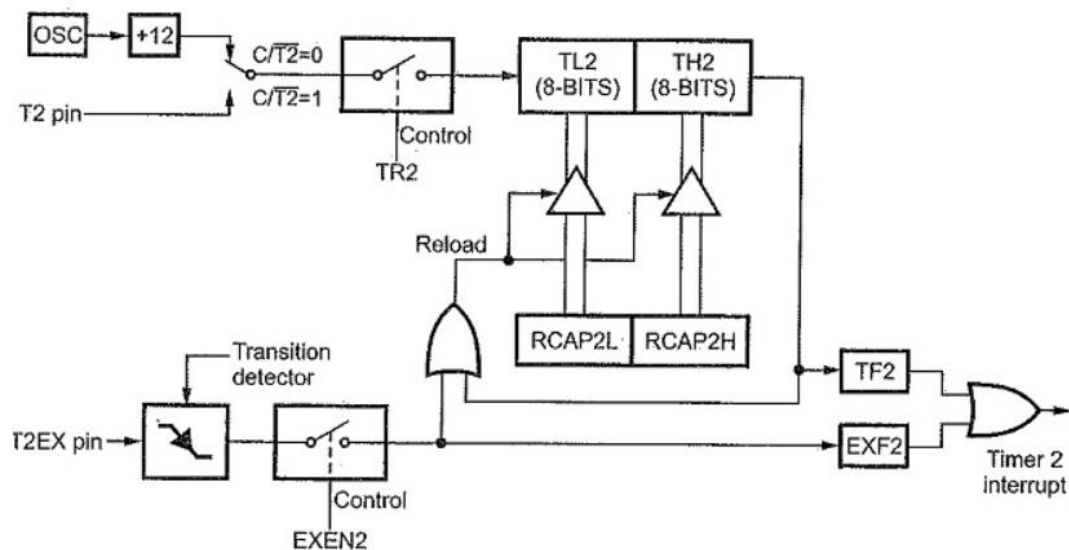
**T2CON – SFR Bit Address**

The operation of Timer 2 (T2) is controlled almost entirely by the T2CON SFR, at address C8h. Note that since this SFR is evenly divisible by 8 that it is bit-addressable.

| BIT | NAME   | BIT ADDRESS | DESCRIPTION                                                                                                                                                                                                      |
|-----|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7   | TF2    | CFh         | <b>Timer 2 Overflow.</b> This bit is set when T2 overflows. When T2 interrupt is enabled, this bit will cause the interrupt to be triggered. This bit will not be set if either TCLK or RCLK bits are set.       |
| 6   | EXF2   | CEh         | <b>Timer 2 External Flag.</b> Set by a reload or capture caused by a 1-0 transition on T2EX (P1.1), but only when EXEN2 is set. When T2 interrupt is enabled, this bit will cause the interrupt to be triggered. |
| 5   | RCLK   | CDh         | <b>Timer 2 Receive Clock.</b> When this bit is set, Timer 2 will be used to determine the serial port receive baud rate. When clear, Timer 1 will be used.                                                       |
| 4   | TCLK   | CCh         | <b>Timer 2 Transmit Clock.</b> When this bit is set, Timer 2 will be used to determine the serial port transmit baud rate. When clear, Timer 1 will be used.                                                     |
| 3   | EXEN2  | CBh         | <b>Timer 2 External Enable.</b> When set, a 1-0 transition on T2EX (P1.1) will cause a capture or reload to occur.                                                                                               |
| 2   | TR2    | CAh         | <b>Timer 2 Run.</b> When set, timer 2 will be turned on. Otherwise, it is turned off.                                                                                                                            |
| 1   | C/T2   | C9h         | <b>Timer 2 Counter/Interval Timer.</b> If clear, Timer 2 is an interval counter. If set, Timer 2 is incremented by 1-0 transition on T2 (P1.0).                                                                  |
| 0   | CP/RL2 | C8h         | <b>Timer 2 Capture/Reload.</b> If clear, auto reload occurs on timer 2 overflow, or T2EX 1-0 transition if EXEN2 is set. If set, a capture will occur on a 1-0 transition of T2EX if EXEN2 is set.               |

**Timer 2 Mode Select**

| RCLK + TCLK | CP/RL2 | TR2 | Mode                |
|-------------|--------|-----|---------------------|
| 0           | 0      | 1   | 16-bit Auto-Reload  |
| 0           | 1      | 1   | 16-bit Capture      |
| 1           | X      | 1   | Baud Rate Generator |
| X           | X      | 0   | (Off)               |

**Timer 2 – 16 Bit Capture Mode****Timer 2 – 16 Bit Auto Reload Mode**

|        |     |         |
|--------|-----|---------|
| T2CON  | EQU | 0C8H    |
| RCAP2L | EQU | 0CAH    |
| RCAP2H | EQU | 0CBH    |
| TL2    | EQU | 0CCH    |
| TH2    | EQU | 0CDH    |
| TR2    | EQU | T2CON.2 |
| TF2    | EQU | T2CON.7 |
| ET2    | EQU | IE.5    |

|            |     |       |
|------------|-----|-------|
| Port_7Seg  | EQU | P0    |
| P_SegLeft  | EQU | P1.4  |
| P_SegRight | EQU | P1.5  |
| Counter    | EQU | 20H   |
| BitDispLR  | EQU | 21H.0 |

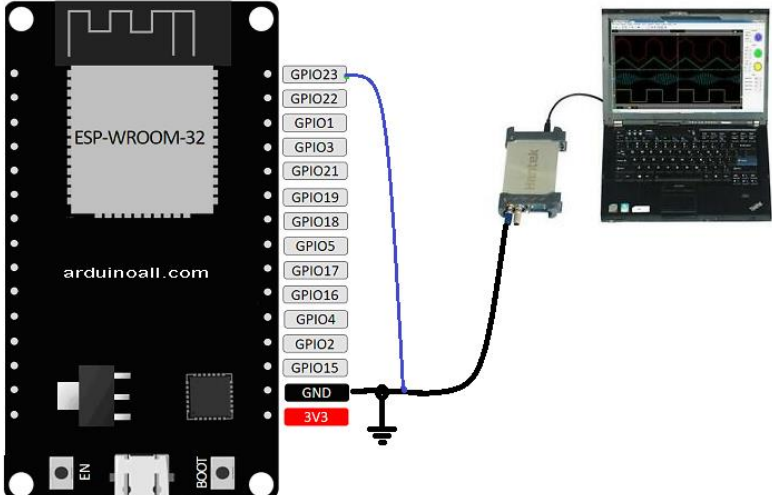
|     |       |
|-----|-------|
| ORG | 0000H |
| JMP | 0100H |

|             |      |                                        |                            |
|-------------|------|----------------------------------------|----------------------------|
|             | ORG  | 002BH                                  | ; Timer2 Interrupt Vector  |
|             | JMP  | T2_ISR                                 |                            |
|             | ORG  | 0100H                                  |                            |
|             | CLR  | EA                                     |                            |
|             | MOV  | SP,#3FH                                |                            |
|             | MOV  | T2CON,#00                              | ; Timer 16Bit Auto-Reload  |
|             | MOV  | DPTR,#0D000H                           | ; number of MC             |
|             | Mov  | RCAP2H,DPH                             |                            |
|             | Mov  | RCAP2L,DPL                             |                            |
|             | Mov  | TH2,DPH                                |                            |
|             | Mov  | TL2,DPL                                |                            |
|             | SETB | TR2                                    | ; Start Timer2             |
|             | SETB | ET2                                    | ; Enable Timer2            |
|             | SETB | EA                                     | ; Enable All               |
| Main_Loop:  | DEC  | Counter                                |                            |
|             | CALL | Delay                                  |                            |
|             | JMP  | Main_Loop                              |                            |
| Delay:      | MOV  | R7,#20                                 |                            |
|             | MOV  | A,#00                                  |                            |
|             | MOV  | B,#00                                  |                            |
| _Dly00:     | DJNZ | B,_Dly00                               |                            |
|             | DJNZ | Acc,_Dly00                             |                            |
|             | DJNZ | R7,_Dly00                              |                            |
|             | RET  |                                        |                            |
| T2_ISR:     | PUSH | Acc                                    | ; Save DPTR, Acc to Stack  |
|             | PUSH | DPH                                    |                            |
|             | PUSH | DPL                                    |                            |
|             | CLR  | TF2                                    | ; Clear Timer2 Overflow    |
|             | MOV  | A,Counter                              |                            |
|             | JB   | BitDisplR,_DRight                      |                            |
| _DLeft:     | SETB | P_SegLeft                              |                            |
|             | CLR  | P_SegRight                             |                            |
|             | JMP  | _DShow                                 |                            |
| _DRight:    | CLR  | P_SegLeft                              |                            |
|             | SETB | P_SegRight                             |                            |
|             | SWAP | A                                      |                            |
| _DShow:     | ANL  | A,#0FH                                 |                            |
|             | MOV  | DPTR,#CODE_TABLE                       |                            |
|             | MOVC | A,@A+DPTR                              |                            |
|             | CPL  | A                                      |                            |
|             | MOV  | Port_7Seg,A                            |                            |
|             | CPL  | BitDisplR                              |                            |
|             | POP  | DPL                                    | ; Get DPTR, Acc from Stack |
|             | POP  | DPH                                    |                            |
|             | POP  | Acc                                    |                            |
|             | RETI |                                        | ; Return from Interrupt    |
| CODE_TABLE: | DB   | 77H, 11H, 6DH, 5DH, 1BH, 5EH, 7EH, 15H |                            |
|             | DB   | 7FH, 5FH, 3FH, 7AH, 66H, 79H, 6EH, 2EH |                            |
|             | END  |                                        |                            |

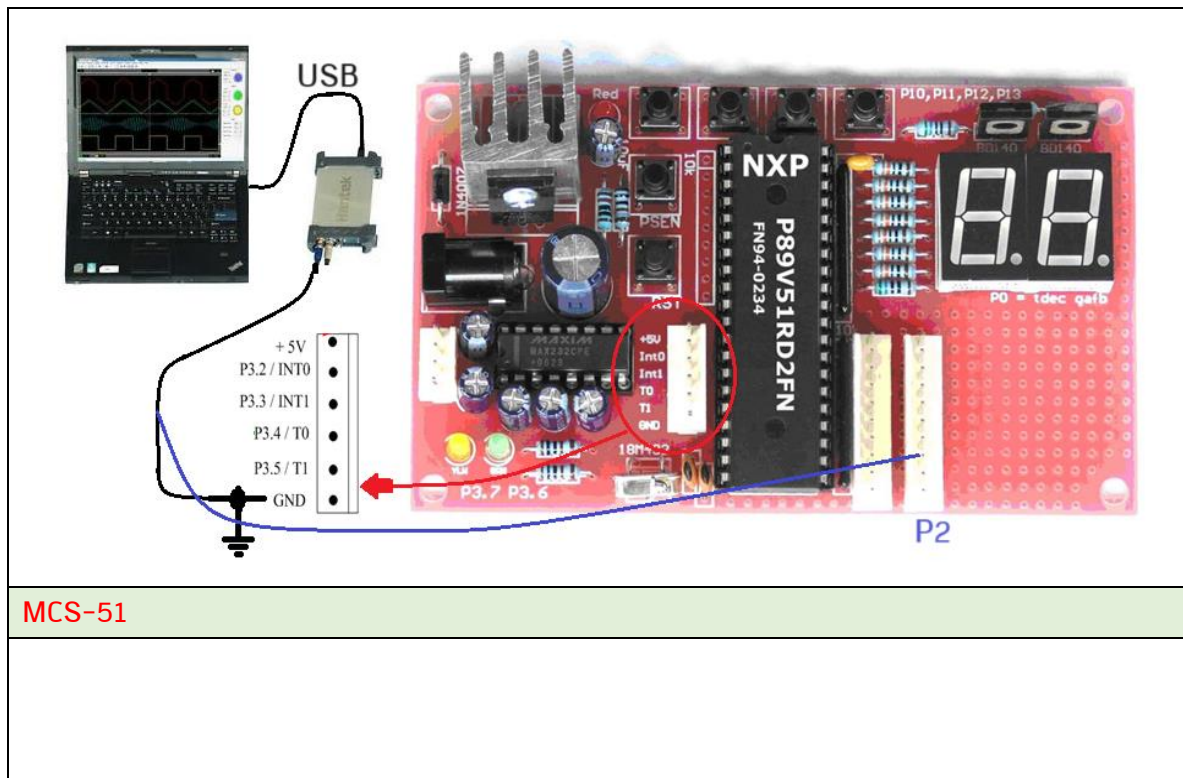
## 2/6: MCS-51 VS ESP-32 in Timer/Counter

### 3A. Timer Delay >> สร้างสัญญาณ → จำนวน, เวลาOn, เวลาOff

3a.1 ใช้ ESP32 สร้างความถี่ 1000 + รหัสสามตัวท้าย ( $\pm 10$ ) Hz หาก รหัสคู่กำหนด  $\tau = 33\%$  และ รหัสคี่กำหนด  $\tau = 66\%$

|                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                     |                                                                                                                                                                                                                                                                                      |
| <b>ESP-32: Classic &lt; 1,033 Hz &gt;</b>                                                                                                                                                                                            | <b>ESP-32: Library &lt; 1,033Hz &gt;</b>                                                                                                                                                                                                                                             |
| <pre>int XX = 1000000 / 1033; // MicroSec void setup() {   pinMode(23, OUTPUT); }  void loop() {   digitalWrite(23, HIGH);   delay(0.66 * XX);   digitalWrite(23, LOW);   delay(0.34 * XX); }</pre> <p><i>on 66%<br/>off 34%</i></p> | <pre>int ledPin = 23; int freq = 1033; int ledChannel = 0; int resolution = 8; // 255 Scale int dutyCycle = 255 * 0.66;  void setup() {   ledcSetup(ledChannel, freq, resolution);   ledcAttachPin(ledPin, ledChannel);   ledcWrite(ledChannel, dutyCycle); }  void loop() { }</pre> |

3a.2 จาก Lab8\_Timer.Asm ให้สร้างควมถี่ 1000 + รหัสสามตัวท้าย ( $\pm 10$ ) Hz รหัสคู่กำหนด  $\tau = 33\%$  และ รหัสคี่กำหนด  $\tau = 66\%$

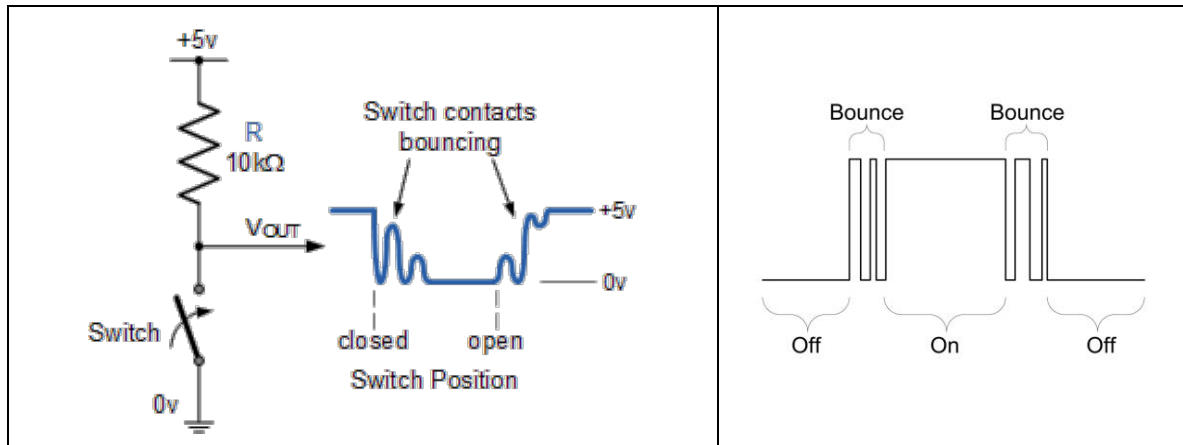


### 3B. Counter with Software >> อ่านสัญญาณ → จำนวน, เวลาOn, เวลาOff

Debounce คืออะไร???

ก่อนจะทำความรู้จัก Debounce นั้นต้องทราบก่อนว่า Bounce คืออะไร

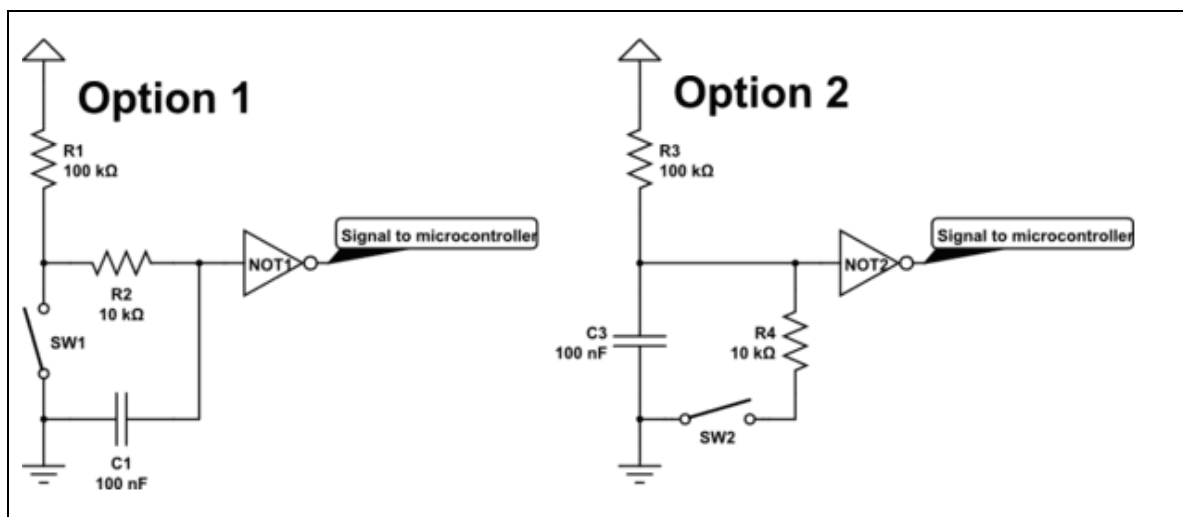
Bounce คือ สัญญาณที่เกิดขึ้นจากการที่การกดปุ่มแต่ละครั้ง เมื่อกดปุ่มหน้าสัมผัสที่ทำการสัมผัสกันนั้น ยังไม่แนบสนิทดี จะทำให้เกิดสัญญาณวิ่งขึ้นวิ่งลง หรือ ก็คือ สัญญาณไม่นิ่ง ซึ่งจะทำให้ค่าที่ส่งออกไปนั้นคงาคเคลื่อน จึงต้องมีการทำ Debounce ขึ้นเพื่อป้องกันเหตุการณ์ที่จะเกิดขึ้นนี้



รูปภาพการเกิด Bouncing

จากรูปจะเห็นได้ว่าช่วง Switch Contact จะเกิดการ Bouncing ที่วงจรเกิดการ เปิด หรือ ปิด ซึ่ง หลักการทำงานของ Debounce คือ จะทำการนับลูกคลื่นของสัญญาณนั้นไปเรื่อยๆ จน สัญญาณนั้นเกิดการนิ่งแล้วซึ่งก็คือตรงกลางของรูปตัวอย่าง เพื่อค่าที่ส่งนั้นจะเกิดความถูกต้องไม่คลาดเคลื่อน

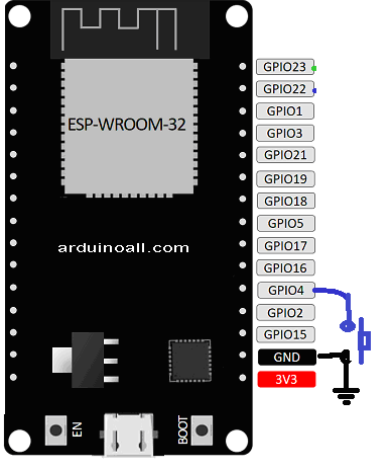
สามารถแก้ไขปัญหาการเกิด Bouncing ได้ด้วยการต่อวงจรภายนอก (Hardware Debounce) และการแก้ด้วยโปรแกรม (Software Debounce)



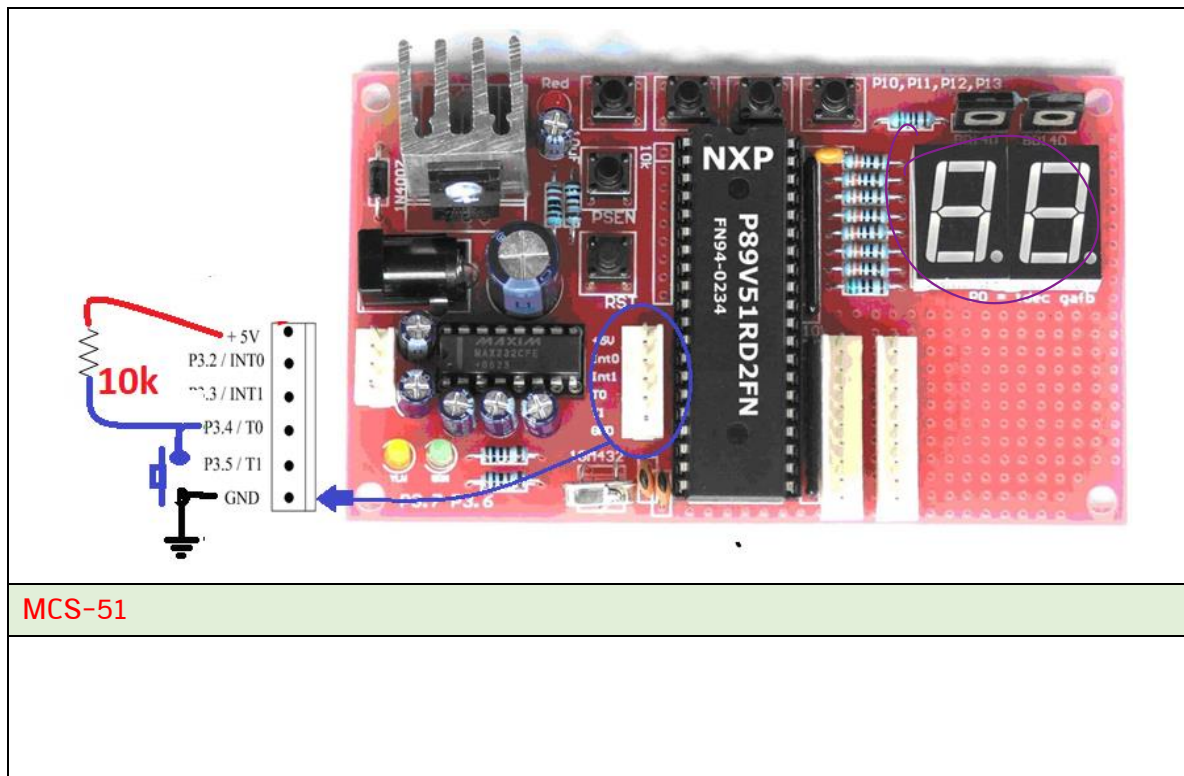
Hardware Debounce

3b.1 ใช้ ESP32 อ่านจำนวนครั้งการกดสวิตช์ แสดงจำนวนครั้งที่ Serial Terminal

<https://poundxi.com/tag/switch-debouncing>

|                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                      | <p><b>ESP-32: Classic</b></p> <pre>int InputPin = 4; unsigned long int XXXX;  void setup() {   pinMode(InputPin, INPUT_PULLUP);   Serial.begin(115200);   XXXX = 0; }  void loop() {   Serial.println(XXXX);   while (digitalRead(InputPin) == 0); delay(1);   while (digitalRead(InputPin) == 1); delay(1);   XXXX++; }</pre> <p><i>Handwritten notes: "50 70" and a square wave diagram.</i></p> |
| <p><b>ESP-321: Library</b></p>                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Add Lib</b></p> <div data-bbox="209 898 1366 1032"> <p><b>ButtonDebounce</b> by Maykon L. Capellari<br/>       A library that makes easy to use button with debounce. Make easy to use button with debounce.<br/> <a href="#">More info</a></p> <p style="text-align: right;"><b>Install</b></p> </div>                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                    |
| <pre>#define pinInput 4 #define DebounceTimeMilliS 1 #include &lt;ButtonDebounce.h&gt;  ButtonDebounce button(pinInput, DebounceTimeMilliS); unsigned long int XXXX;  void buttonChanged(int state) {   if (button.state() == LOW) Serial.println(++XXXX); }  void setup() {   Serial.begin(115200);   button.setCallback(buttonChanged); }  void loop() {   button.update(); }</pre> |                                                                                                                                                                                                                                                                                                                                                                                                    |

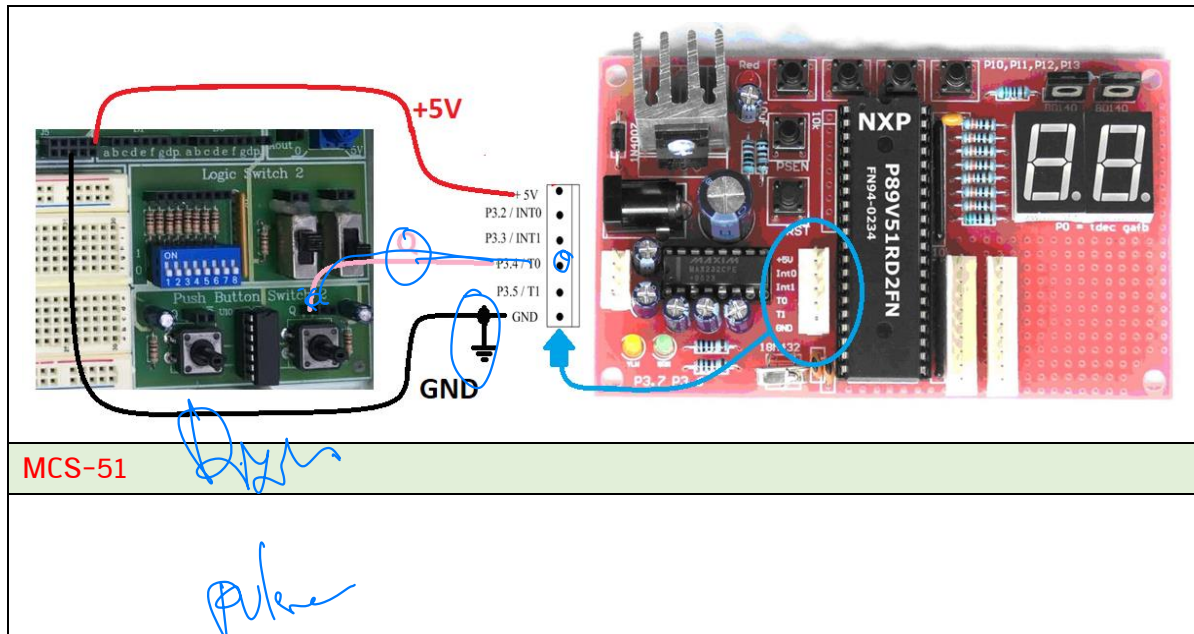
3b.2 จาก Lab8\_Counter1.Asm อ่านจำนวนครั้งการกดสวิตช์ แสดงผลที่ Left Segment



### 3C. Counter with Hardware >> อ่านสัญญาณ → จำนวน, เวลาOn, เวลาOff

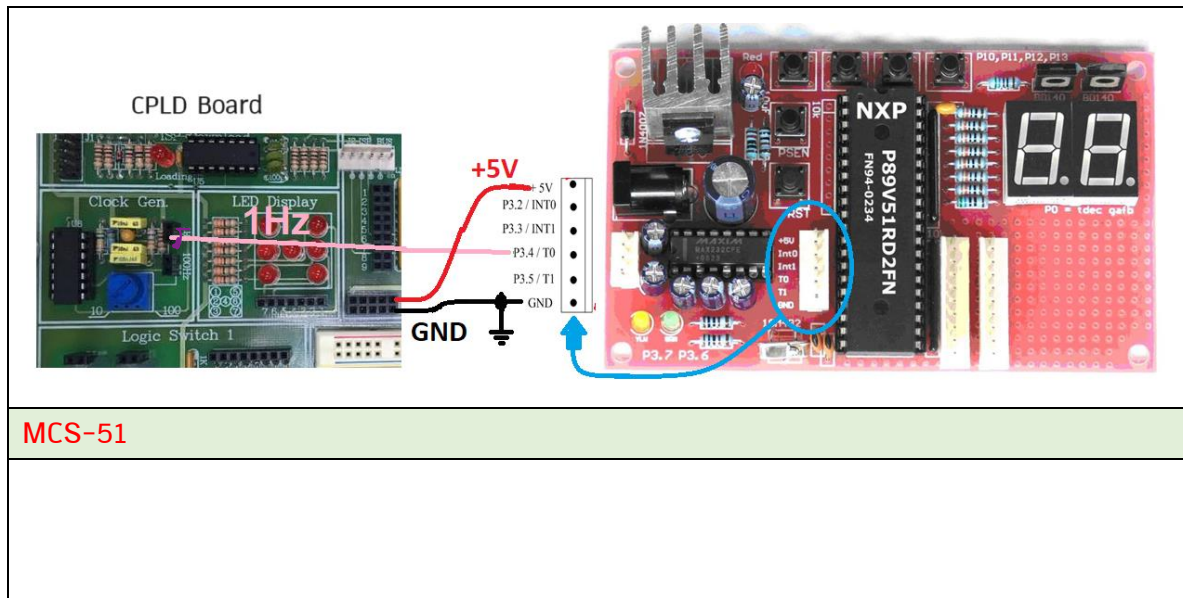
3C.1 ให้เขียนโปรแกรมตาม Lab8\_Counter2.Asm เป็นโปรแกรมสำหรับการนับค่าด้วยการใช้ Counter0 เพื่อ นับสัญญาณที่ขา P3.4(TO) ทดสอบด้วยการแตะสัญญาณ GND ที่ขา TO สังเกตผลการนับที่ได้

- หากต้องการรับสัญญาณที่ขา T1 จะปรับปรุงแก้ไขโปรแกรมอย่างไร
- ระหว่างการป้อนอินพุต แบบข้อ 3b.2, ตามรูป Q, ตามรูปแต่เป็น Q-bar ผลการทำงาน ต่างกันอย่างไร เพราะเหตุใด



Deban Supt d

3C.2 ให้เขียนโปรแกรมตาม Lab8\_Counter3.Asm เป็นโปรแกรมสำหรับการนับค่าด้วยการใช้ Counter0 เพื่อนับสัญญาณที่ขา P3.4(T0) และแสดงผลการนับ TH0-TL0 ที่ 7\_Segment 2หลัก ทดสอบด้วยการป้อน Clock 1Hz จาก CPLD Board ที่ขา T0 สังเกตผลการนับที่ได้



### 3D. RIDE-51 C for MCS-51

```
//=====
// Freq 1,033 Hz ==> Period = 968.0542 uSec
//                               ==> Duty 50% = 484.0271 uSec
// XTAL : 18.4322 MHz
// 1 MC = 12/18.432MHz = 0.651041667 uSec --> Normal Mode
// 484.0271 uSec = 743.465634 MC --> 744 MC
// 65536-744 = 64792 = 0xFD18
//
// 1 MC = 6/18.432MHz = 0.3255208333 uSec --> x2 Mode
// 484.0271 uSec = 1486.93127 MC --> 1487 MC
// 65536-1487 = 64049 = 0xFA31
//
//=====
#include <REG51.H>
void main()                // Main
{   TMOD = 0x01;           // Timer 16 bit
    while(1)
    {   TR0 = 0;
        TH0 = 0xFA;
        TL0 = 0x31+12;      // +12 for move & compare instruction
        TF0 = 0;
        TR0 = 1;
        while(TF0 == 0);
        P2 ^= 0x0ff;
    }
}
```

#### 3d.1 Send Pulse Program using RIDE-C51

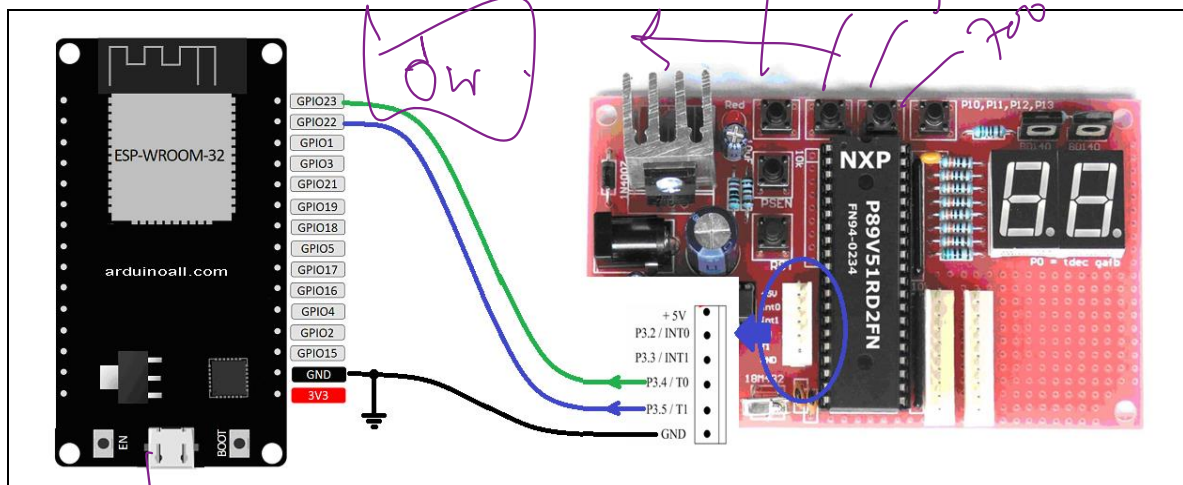
- ให้เขียนโปรแกรมด้วยภาษา C เพื่อรับค่าจาก P1 กำหนดให้ การทำงานตามตาราง
- ทดสอบการทำงานด้วยการป้อนอินพุตให้กับวงจรในข้อ 3C2
- หากส่ง Pulse ครบจำนวนที่กำหนดแล้ว ให้รอการกด Switch ใหม่ และให้ P3.2=High, 7Segment แสดง --

| มีการกด Switch | จะทำให้ | 7_Seg แสดง | จำนวน Pulse ที่ส่งออกที่ P3.2 | ระยะห่างแต่ละ Pulse |
|----------------|---------|------------|-------------------------------|---------------------|
| SW0            | P1.0=0  | 11         | 100                           | 12 mSec             |
| SW1            | P1.1=0  | 33         | 300                           | 12 mSec             |
| SW2            | P1.2=0  | 55         | 500                           | 6 mSec              |
| SW3            | P1.3=0  | 77         | 700                           | 6 mSec              |

**MCS-51**

### 3E. Pulse from MCS-51 to ESP-32

3E.1 ให้ MCS51 ส่ง pulse ออกที่ขา P3.4, P3.5 แล้วให้ ESP32 อ่านจำนวน Pulse ที่ส่งออกมา ให้แสดงค่าที่ได้ที่ Serial Terminal



#### ESP-32: Pulse Count

- Example: for 1 Input D23(ยังไม่สมบูรณ์ อยากได้ 2 Input D22, D23)>
- <https://www.arduinoall.com/article/164/11-esp32-สอน-arduino-esp32-การใช้งาน-arduino-esp32-interrupt>

```
int xPin = 23;
int mPin = 0;
unsigned long int XXXX = 0;
```

```
void setup() {
  pinMode(xPin, INPUT_PULLUP);
  pinMode(mPin, INPUT_PULLUP);
  Serial.begin(115200);
  Serial.println("\n\nPress Boot_Sw for Show Counter");
  XXXX = 0;
}
```

```
void loop() {
  while (digitalRead(xPin) == 1) if (digitalRead(mPin) == 0) Serial.println(XXXX);
  while (digitalRead(xPin) == 0) if (digitalRead(mPin) == 0) Serial.println(XXXX);
  XXXX++;
}
```

## MCS-51: Pulse Out

```

AP_DLY      EQU      30H
AN_DLY      EQU      31H
ANP_H       EQU      32H
ANP_L       EQU      33H
BP_DLY      EQU      34H
BN_DLY      EQU      35H
BNP_H       EQU      36H
BNP_L       EQU      37H

                ORG      0000H
                JMP      0100H

                ORG      0100H
                MOV      SP,#3FH
                CLR      EA
                MOV      AP_DLY,#12      ; delay Step
                MOV      DPTR,#-200     ; nPluse x2
                MOV      ANP_H,DPH
                MOV      ANP_L,DPL
                MOV      BP_DLY,#34     ; delay Step
                MOV      DPTR,#-600    ; nPluse x2
                MOV      BNP_H,DPH
                MOV      BNP_L,DPL

                MOV      AN_DLY,AP_DLY
                MOV      BN_DLY,BP_DLY
LOOP:         CALL      DLY
                CALL      PA_CHK
                CALL      PB_CHK
                JMP      LOOP

;=====
PA_CHK:      MOV      A,ANP_H
                JNZ      _DC_NA
                MOV      A,ANP_L
                JNZ      _DC_NA
                RET
_DC_NA:      DEC      AN_DLY
                MOV      A,AN_DLY
                JNZ      _XPACK
                MOV      DPH,ANP_H
                MOV      DPL,ANP_L
                INC      DPTR
                MOV      ANP_H,DPH
                MOV      ANP_L,DPL
                CPL      P3.4
                MOV      AN_DLY,AP_DLY
_XPACK:      RET

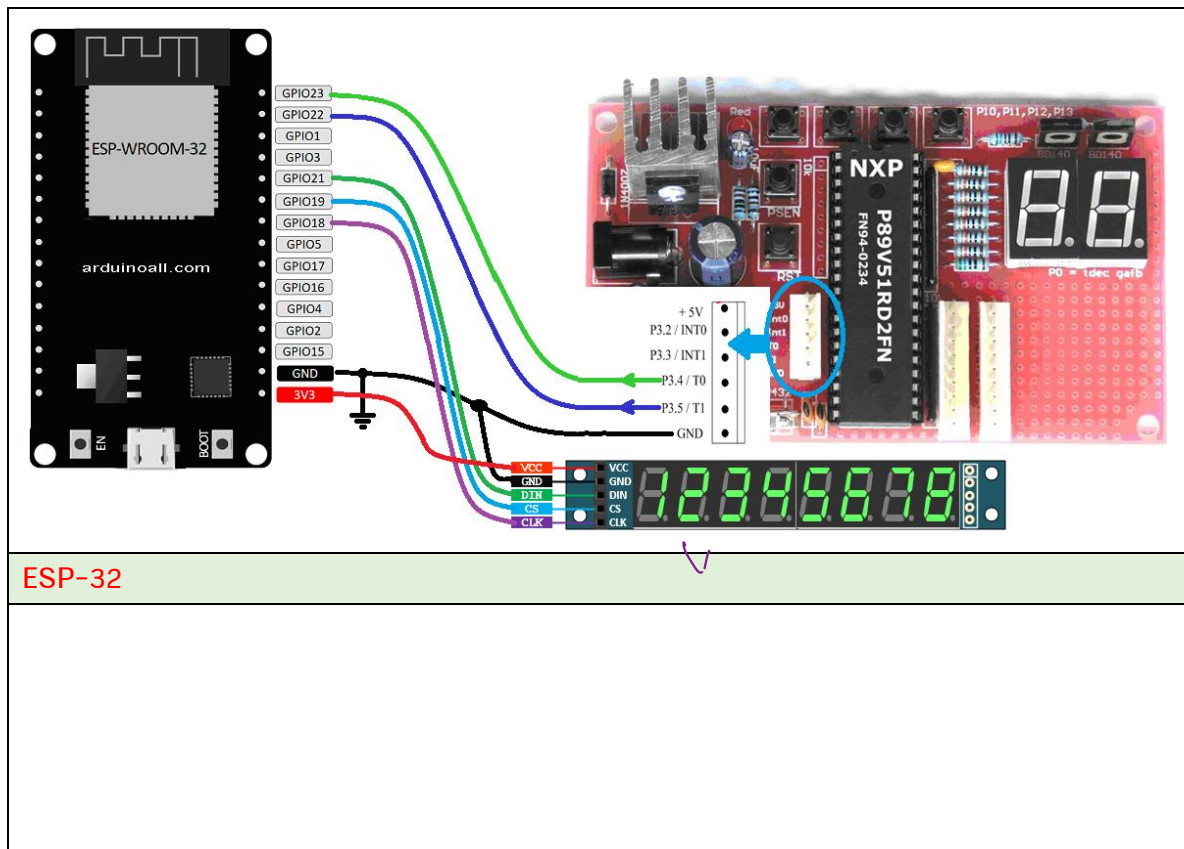
;=====
PB_CHK:      MOV      A,BNP_H
                JNZ      _DC_NB
                MOV      A,BNP_L
                JNZ      _DC_NB
                RET
_DC_NB:      DEC      BN_DLY
                MOV      A,BN_DLY
                JNZ      _XPBCK
                MOV      DPH,BNP_H
                MOV      DPL,BNP_L
                INC      DPTR
                MOV      BNP_H,DPH
                MOV      BNP_L,DPL
                CPL      P3.5
                MOV      BN_DLY,BP_DLY
_XPBCK:      RET

;=====
DLY:         MOV      A,#80H
                DJNZ      Acc,$
                RET

                END

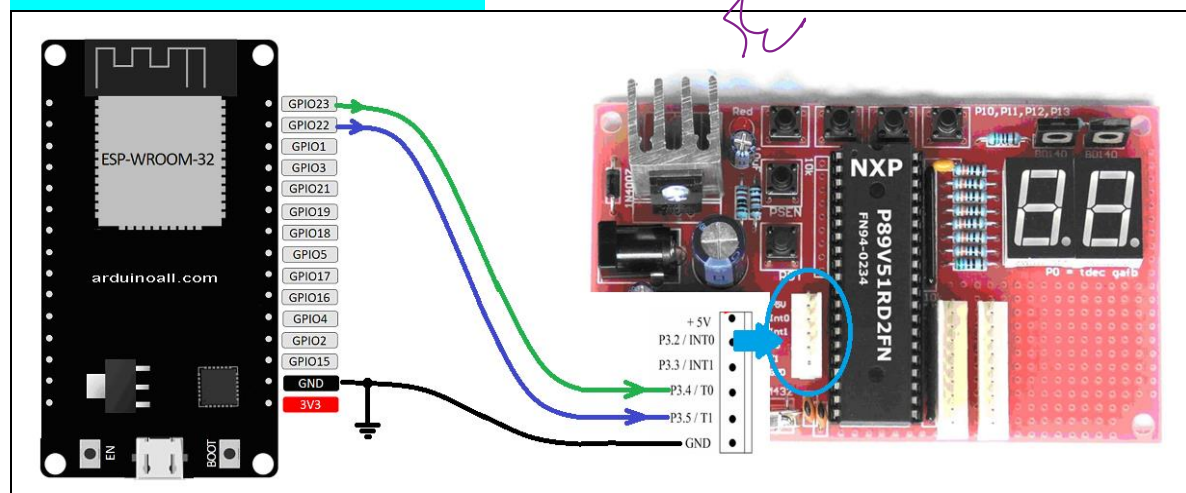
```

3E.2 จาก 3E.1 นอกจากแสดงค่าที่ได้ที่ Serial Terminal แล้วให้แสดงค่าที่ MAX7219 Display



Tx

## 3F. Pulse from ESP-32 to MCS-51



ให้ ESP32 ส่ง pulse ออกที่ขา 22, 23 แล้วให้ MCS51 อ่านจำนวน Pulse ที่ส่งออกมา

- ให้แสดงค่าที่ได้จากขา 22 จำนวน 2 หลักที่ 7Segment เมื่อกด SW\_P1.0
- ให้แสดงค่าที่ได้จากขา 23 จำนวน 2 หลักที่ 7Segment เมื่อกด SW\_P1.3

## ESP-32

```
int Pin_A = 22; // Pin Assignment
int Pin_B = 23;
```

```
int nPulse_A = 0x123; // number of pulse
int nPulse_B = 0x456;
int dPulse_A = 43; // pulse delay (ms)
int dPulse_B = 21;
```

```
unsigned long int nCounter = 0;
int stsPulse_A = 0;
int stsPulse_B = 0;
```

```
void setup() {
  pinMode(Pin_A, OUTPUT);
  pinMode(Pin_B, OUTPUT);
  digitalWrite(Pin_A, HIGH);
  digitalWrite(Pin_B, HIGH);
  nPulse_A = nPulse_A * 2; // On and Off
  nPulse_B = nPulse_B * 2;
}
```

```
void loop() {
  delay(1);
  nCounter++;
  if (((nCounter % dPulse_A) == 0) && (nPulse_A >= 0)) {
    stsPulse_A = 1 - stsPulse_A;
    digitalWrite(Pin_A, stsPulse_A);
    nPulse_A--;
  }
  if (((nCounter % dPulse_B) == 0) && (nPulse_B >= 0)) {
    stsPulse_B = 1 - stsPulse_B;
    digitalWrite(Pin_B, stsPulse_B);
    nPulse_B--;
  }
}
```

## MCS-51

```

T2CON      EQU    0C8H
TH2        EQU    0CDH
TL2        EQU    0CCH
RCAP2H     EQU    0CBH
RCAP2L     EQU    0CAH
TR2        EQU    T2CON.2
TF2        EQU    T2CON.7
ET2        EQU    IE.5

Port_7Seg  EQU    P0
P_SegLeft  EQU    P1.4
P_SegRight EQU    P1.5
P_SW0      EQU    P1.0
P_SW3      EQU    P1.3
P_T0_34    EQU    P3.4
P_T1_35    EQU    P3.5
Counter    EQU    20H
BitDisplR  EQU    21H.0
BitShowAB  EQU    21H.1

                ORG    0000H
                JMP    0100H

                ORG    002BH
                JMP    T2_ISR

                ORG    0100H
                CLR    EA
                MOV    SP,#3FH
                SETB   P_SW0
                SETB   P_SW3
                SETB   P_T0_34
                SETB   P_T1_35

                MOV    TMOD,#01010101B
                MOV    TH0,#00
                MOV    TL0,#00
                MOV    TH1,#00
                MOV    TL1,#00
                SETB   TR0
                SETB   TR1

                MOV    T2CON,#00
                ; Timer 16Bit Auto-Reload
                MOV    DPTR,#0D000H
                MOV    RCAP2H,DPH
                MOV    RCAP2L,DPL
                MOV    TH2,DPH
                MOV    TL2,DPL
                SETB   TR2      ; Start Timer2
                SETB   ET2      ; Enable Timer2
                SETB   EA      ; Enable All

                MOV    Counter,TL0

```

```

Main_Loop:  NOP
_Chk_SW0:   JB     P_SW0,_Chk_SW3
            SETB   BitShowAB

_Chk_SW3:   JB     P_SW3,_Chk_Show0
            CLR    BitShowAB

_Chk_Show0: JB     BitShowAB,_Chk_Show1
            MOV    Counter,TL0
            JMP    Main_Loop

_Chk_Show1: MOV    Counter,TL1
            JMP    Main_Loop

Delay:      MOV    R7,#20
            MOV    A,#00
            MOV    B,#00
_Dly00:     DJNZ   B,_Dly00
            DJNZ   Acc,_Dly00
            DJNZ   R7,_Dly00
            RET

T2_ISR:     PUSH    Acc
            PUSH    DPH
            PUSH    DPL
            CLR    TF2
            MOV    A,Counter
            JB     BitDisplR,_DRight
_DLeft:     SETB   P_SegLeft
            CLR    P_SegRight
            JMP    _DShow
_DRight:    CLR    P_SegLeft
            SETB   P_SegRight
            SWAP    A
_DShow:     ANL    A,#0FH
            MOV    DPTR,#CODE_TABLE
            MOVC   A,@A+DPTR
            CPL    A
            MOV    Port_7Seg,A
            CPL    BitDisplR
            POP     DPL
            POP     DPH
            POP     Acc
            RETI

CODE_TABLE: DB     77H, 11H, 6DH, 5DH
            DB     1BH, 5EH, 7EH, 15H
            DB     7FH, 5FH, 3FH, 7AH
            DB     66H, 79H, 6EH, 2EH

            END

```

### 3/6: ESP-32: RTOS

- <https://www.iot.eng.kmutnb.ac.th/pub/docs/>
- <http://www.ayarafun.com/2016/12/basic-freertos-with-node32s-esp32/>
- <https://attaphon.medium.com/ลองเขียนโปรแกรมแบบ-multi-tasking-บน-esp32-กัน-rtos-fd098f798f8f>
- <https://www.hackster.io/luo-zeqing/implement-freertos-with-arduino-ide-on-esp32-444f54>

#### ESP-32

```
// -----
#define BUILDIN_LED 2
#define ledPIN1 23
#define ledPIN2 22
int delayTask1 = 50, delayTask2 = 500; // mSec

// -----
void first_task(void *pvParameter) {
    pinMode(ledPIN1, OUTPUT);
    while (true) {
        digitalWrite(ledPIN1, HIGH);
        vTaskDelay(delayTask1 / portTICK_RATE_MS);
        digitalWrite(ledPIN1, LOW);
        vTaskDelay(delayTask1 / portTICK_RATE_MS);
    }
}

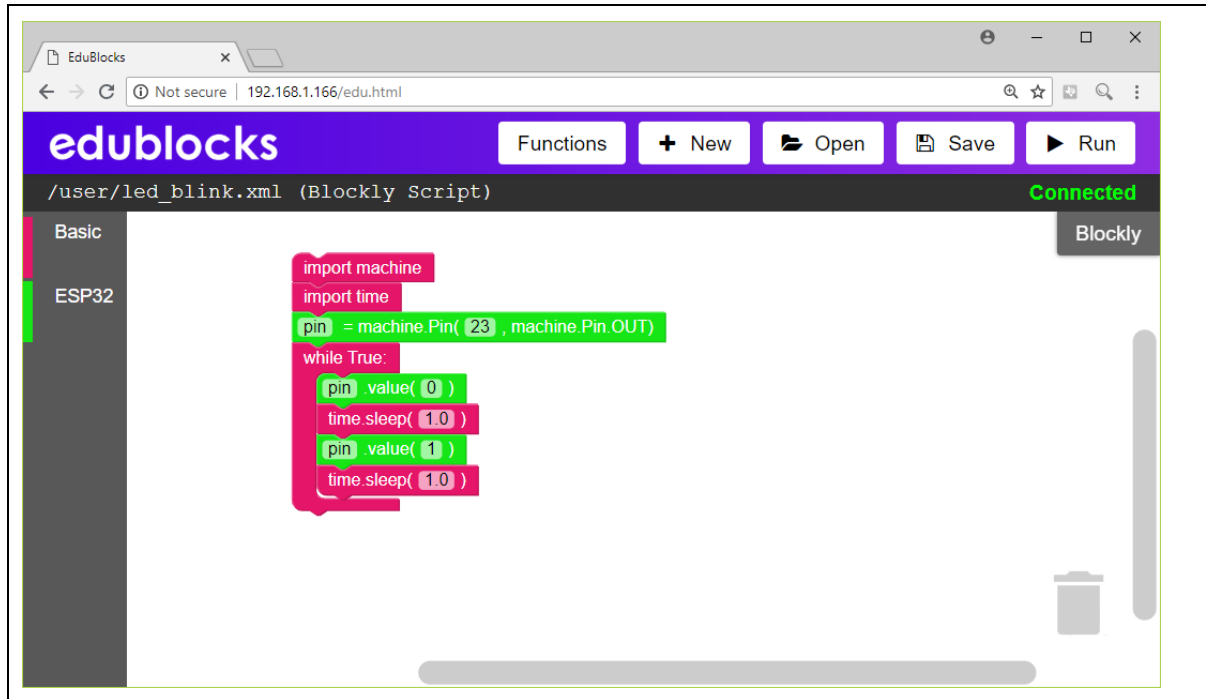
// -----
void second_task(void *pvParameter) {
    pinMode(ledPIN2, OUTPUT);
    while (true) {
        digitalWrite(ledPIN2, HIGH);
        vTaskDelay(delayTask2 / portTICK_RATE_MS);
        digitalWrite(ledPIN2, LOW);
        vTaskDelay(delayTask2 / portTICK_RATE_MS);
    }
}

// -----
void setup() {
    Serial.begin(115200);
    pinMode(BUILDIN_LED, OUTPUT);
    // xTaskCreate(pvTaskCode, pcName, usStackDepth, pvParameters, uxPriority, pxCreatedTask)
    xTaskCreate(&first_task, "first_task", 1024, NULL, 5, NULL);
    xTaskCreate(&second_task, "second_task", 1024, NULL, 5, NULL);
}

// -----
void loop() {
    if (Serial.available() > 0) {
        int temp = Serial.parseInt();
        Serial.println(temp);
        delayTask1 = (temp > 10 & temp < 20) ? 1000 / ((temp - 10) * 2) : delayTask1;
        delayTask2 = (temp > 20 & temp < 30) ? 1000 / ((temp - 20) * 2) : delayTask2;
    }
    digitalWrite(BUILDIN_LED, HIGH);
    delay(250);
    digitalWrite(BUILDIN_LED, LOW);
    delay(250);
}
```

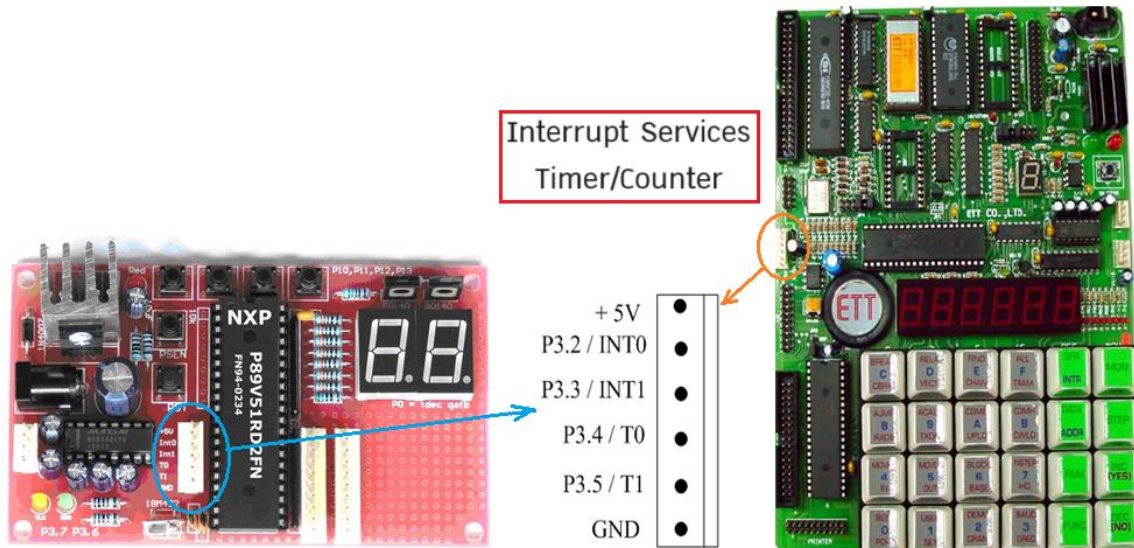
## 4/6: ESP-32: EDUBlocks and MicroPython

- <https://www.electromaker.io/tutorial/blog/welcome-to-the-esp32-running-edublocks>
- [https://www.iot.eng.kmutnb.ac.th/pub/docs/2018/edublocks\\_esp32\\_micropython/](https://www.iot.eng.kmutnb.ac.th/pub/docs/2018/edublocks_esp32_micropython/)

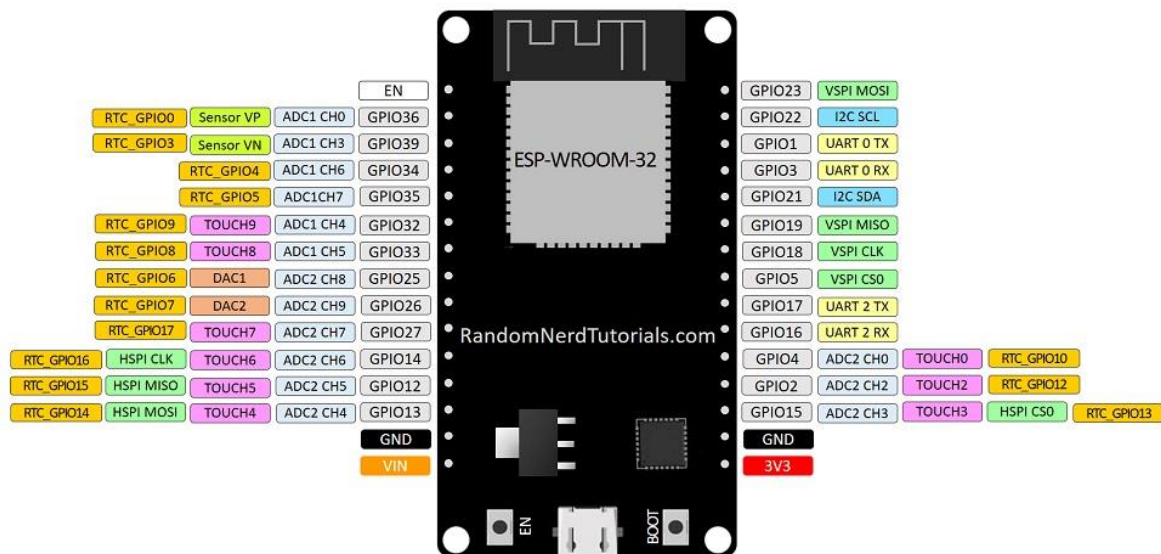


## 5/6: Hardware Reference

## MCS-51: SUT V5 Board, ET-8032 V2 Board



## ESP32 Devkit V1 – DOIT Version with 30 GPIOs



- <https://www.sparkfun.com/tutorials/348>
- <http://forum.arduino.cc/index.php?topic=22060.0>

## 6/6: Node-RED

### Read More

- <http://noderedguide.com/>
- Install and First Flows
  - <https://nodered.org/docs/tutorials/first-flow>
  - <https://www.youtube.com/watch?v=jqJ-D41SFL8>
  - [https://www.youtube.com/watch?v=rQuyjF2\\_Q4M](https://www.youtube.com/watch?v=rQuyjF2_Q4M)
- Install <https://jackrobotics.me/node-red-ep0-install-windows-1c836a715fe>
- Get LINE Token <https://jackrobotics.me/แจ้งเตือน-line-notify-เตือนง่ายได้ใจ-7328125083ec>
- Data to LINE with node-red <https://jackrobotics.me/จับ-line-notify-ยึด-node-red-โดยใช้ความสามารถเดิมของ-node-red-1730d83c067c>

### Experiment

1. Install Node-RED and Start
2. Mission-1: Creating your first flow
3. Mission-2: Node-RED Publish and Subscribe
4. Mission-3: LINE Test >> <https://www.youtube.com/watch?v=2vGtXZRIgy8>
5. Mission-4: LINE Bot with Node-RED >> <https://youtu.be/4KKIEmWYCGQ?t=794>
6. Mission-5: From ESP-32 to MQTT Broker and If xxx then LINE (or Telegram) Alert
  - หากอุณหภูมิ เกิน xxxx หรือ ความชื้น เกิน yyyy ให้มีการแจ้งเตือนผ่าน LINE
7. Mission-6: From ESP-32 to MQTT Broker and Show on Node-RED Dashboard
  - แสดงอุณหภูมิและความชื้นจาก DHT-22 ที่ Node-RED Dashboard
  - ควบคุม LED 2 ตัว